MILITARY SPECIFICATION

FIRE CONTROL COMPUTER, FIELD TEST SET (FCC-FTS)

This specification is approved for use with the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope* This specification contains performance and environmental requirements, quality assurance provisions, and packaging for delivery for the Field Test Set, Fire Control Computer (FCC-F'TS) : M141, which shall be used for the functional test and fault isolation on the Fire Control Computer Digital: M22 (FCC)e The Test Set provides all the interconnecting cables, controls, and displays required to power and test the M22 FCC. The Test Set includes self-test capability.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materials, General Specification Governing the Manufacture and Inspection of
MIL-T-28800 - Test Equipment for use with Electronic and Electrical Equipment
MIL-M-38510 - Microcircuits, General Specification for

Beneficial comments (recommendation, additions,deletions) and any pertinent data which may be of use in improving this document should be addressed in : Commander, U.S.ArmyArmament, Munitions and Chemical Command, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 using the self-addressed Standardization Documentlmprovemerit Propoaal(DDForm 1428)appearing at the end of this document or by letter.

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MIL-s-45743 - Soldering, Manual Type, High Reliability Electrical, Electronic foz Aerospace and Control Systems, Procedure for
ML-S-46844 - Solder Bath. Soldering and Printed Wiring Assemblies, Automatic Machine Type

STANDARDS

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-781 - Reliability Design Qualification and Production Acceptance Tests
MIL-STD-81O - Environmental Test Method for Aerospace and Ground Equipment
MIL-STD-45662 - Calibration System Requirements

2.1.2 Other government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

us ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

11781400 - Test Set, Fire Control Subsystem: M141
11781435 - W1 Cable Assembly (Test)
11781429 - W2 Cable Assembly (Pwr In)
11781401 - W3 Cable Assembly (PCC Pwr)
11781402 - W4 Cable Assembly (FCC 1/0)
11781403 - W5 Cable Assembly (W/A)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. The issues of documents which have not been adopted shall be those in effect on the date of the cited DODISS.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

AsTM_D3951 " Standard Practice for Commercial Packing

TECHNICAL MANUAL

TM9-4931-375-13&P

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Definition. The Test Set shall provide control of the FCC processor, input/output testing of the FCC, and disPlaY of the test results. The Test **Set** shall determine the operability status of the Fcc under test and provide fault detection and isolation of an FCC failure to a Shop Replaceable Unit (SRU). The TeSt set shall have a self test capability.

3.2 Characteristics.

3.2.1 Performance. The Test Set shall provide a control and display panel for operator interface to perform the,FCC diagnostics. The Test Set shall provide input/output stimulation via direct interface and wrap-around cables (connecting FCC digital and analog outputs to inputs), a 400 Hz reference signal source, and resident diagnostic software. An instruction manual, TM-9-4931-375-13, will also be employed by the operator to achieve fault isolation. The Test Set shall provide the operator with a minimum probability of 95% of detecting and isolating an FCC Failure to the Shop Replaceable Unit. Self test of the Test Set shall provide validation of all controls, displays, and resident FCC diagnostic software (via check-sum techniques) with a minimum probability of 80%.

3.2.2 Interface definition.

3.2.2.1 Control interface signals. The Test Set shall have access **to the following computer signal lines to activate the TEST RESULTS display and control diagnostic program, as well as test mode** and selection.

a. zB **Bus** 16 **bit interval Central** Processing Unit (CPU) bus, switches at computer clock rate 3.36 MHz, **carries output data word during output instruction execution** and memory references.

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be Phases. Computer phases" "are Utilized" **to control"** Memory and Display.

c. ZC Bus. 16 bit interval CPU bus, switches at computer clock rate of 3.36 MHz, carries output address during output instruction execution and memory references.

d. INSTR Lines. INSTR Lines are decoded to load display data and 1/0 start condition.

3.2.2.2 FCC input signals. Test Set shall provide as inputs to the FCC the following:

a. 12 bits of code from the TEST SELECT switches and MODE switches.

b. Simulated serial 1/0 signals (Air Data Subsystem and Doppler).

3.2.2.3 Test points interface signals. The following signals are to be available at the test points:

CLOCK	- 3.36 MHz CPU logic clock
PHA, PHB, PHC	CPU Phase (state) counter, time reference for ZB and ZC register monitor
I NSTR	- Instruction Lines
RINTO	- Real Time Interrupt Override
ON	- Power supply sequencing output
MC INT	Memory Cycle Initiate Pulse
MCS, CLD	- CPU bus control lines
HALT	- Programmed **clock stop** command
1/0 SIGNALS	
Power supply voltage monitors	± 15v, +5.5v +5 VDc

3.2.2.4 Test points. **Jacks to observe test points in the event** that the FCC processor is **so disabled that it cannot execute any** programs at all. These test points shall include:

a. **All power supply output voltages.**

b. **Basic system timing - logic and real time clocks.**

c. Critical control functions - memory cycle initiate, memory mode, and power supply logic sequencing signals.

d. Ground returns.

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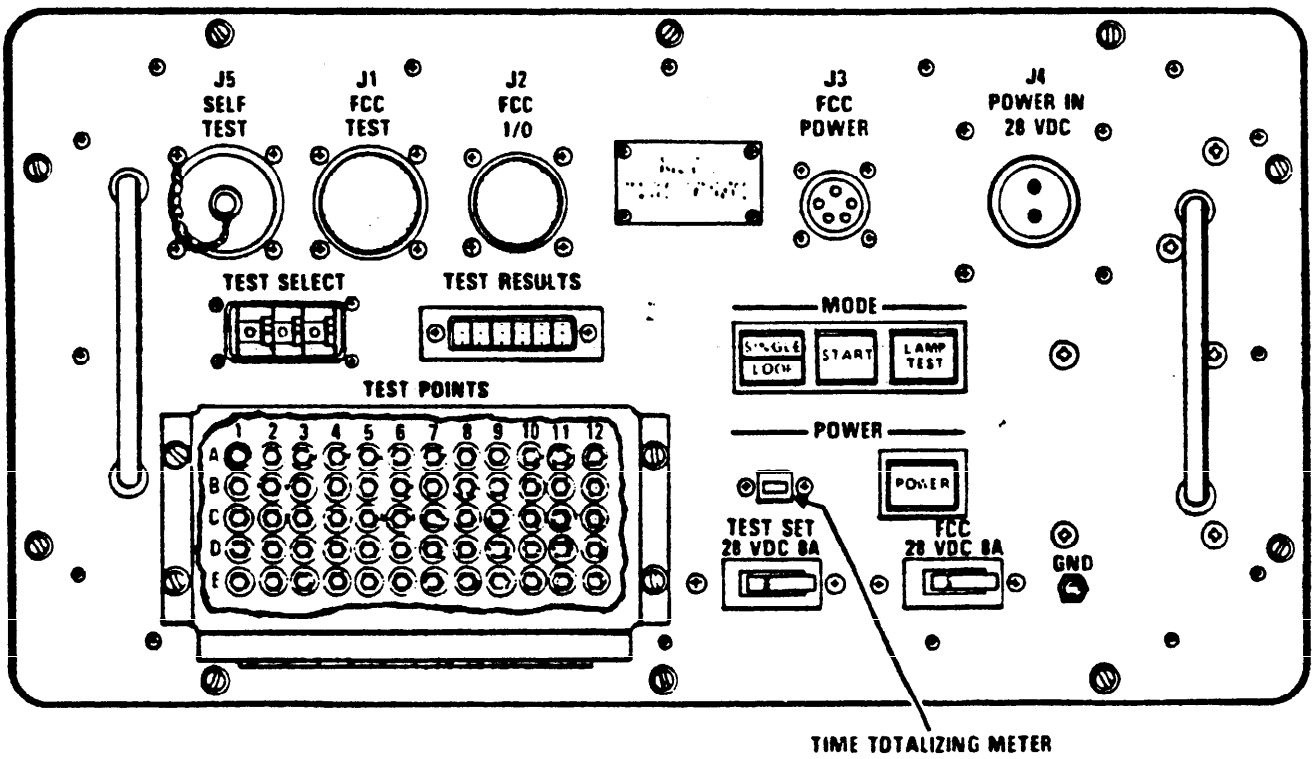


FIGURE 1. Test set control panel

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3. 2.2.5 Connectors. The panel contains 5 main connectors, these are:

a. Power IN. Connects external 28 Volt input power to FCC. J4

b. FCC **TEST. Connects control, display, and test signals** between **Fcc-FTS and FCC. J1**

c. FCC POWER. Connects the +28 volt input power from the F**CC-** FTS to the FCC. J3

d. **FCC** 1/0. Connects digital 1/0 interface signals between FCC-FTS and FCC. J2

e. SELF'-TEST. Connects Self-Test logic to the FCC TEST and FCC 1/0 Connectors. J5

3.4.3 Major commnent characteristics.

3.2.3.1 Combination case. combination case shall be provided to house the Test Set compnents and protect them against adverse environmental conditions. The case provides storage for the instruction manual and Test Set cables.

3.2.3.2 Control panel. The control panel contains the **controls and** displays which perform the functions as specified in paragraph 3.2.1. The control panel layout shall be shown in Figure 1. Control and display functions are as follows:

TEST SELECT switches. Octal thumbwheel switches to provide for up to 512 programmed diagnostic routines which can be executed by the FCC processor.

b. TEST RESULTS display. Six 7-segment indicators to display octal codes representing test results.

c. **SINGLE/LOOP** mode switch. **Alternate action switch which controls whether the test routine will be executed once or repetitively to isolate faults that are intermittent in nature. The mode selected shall be illuminated on the switch.**

d. **START** mode switch. When depressed and released, instructs the program to begin the test routine selected on the **TEST** SELECT thumbwheel **switches** and remains illuminated while the routine is being **executed.**

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LAMP TEST'. When depressed, lights the TEST RESULTS display to all "8's", and illuminates all **Mode** Switches.

f. **POWER switch**. **When depressed lights the POWER indicator, applies power to the FCC under test through the FCC POWER Input connector, and applies power to the test set.**

g. **Circuit breakers**. **Provide +28 VDC overcurrent protection to the Test Set and the FCC.**

h. Time totalizing meter. Indicates accumulated operating time of the Test Set.

3.2.3.3 Test cable. The test cable connects the test set to the FCC test connector.

3.2.3.4 FCC 1/0 wrap-around cable. This cable connects the FCC digital output to FCC digital inputs and FCC analog outputs to FCC analog inputs for test stimulation.

3.2.3.5 Power input cable. This cable connects +28 VDC input power from the test facility to the Test Set.

3.2.3.6 FCC power cable. This cable connects switched +28 VDC power from the Test Set to the FCC power input connector.

3.2.3.7 Self-test wrap-around cable. This cable connects the Test Set self-test connector to the FCC test and FCC 1/0 connectors on the Test Set.

3.2.3.8 Instruction manual. This manual contains operator instructions, test sequences (including self-test), and TEST RESULTS display codes and explanations.

3.2.4 Major component list. The major components of the Test Set are as follows:

Case, Combination
Panel Assembly
W Cable Assembly **(Test)**
w2 Cable Assembly **(Pwr In)**
w3 Cable Assembly **(FcC Pwr)**
w4 Cable Assembly **(FCC 1/0)**
w5 Cable Assembly **(w/A)**
Logic Assembly
Bezel Assembly, Display
400 Hz Ocillator Assmbly
Substructure Assembly
Component Board, Power Control

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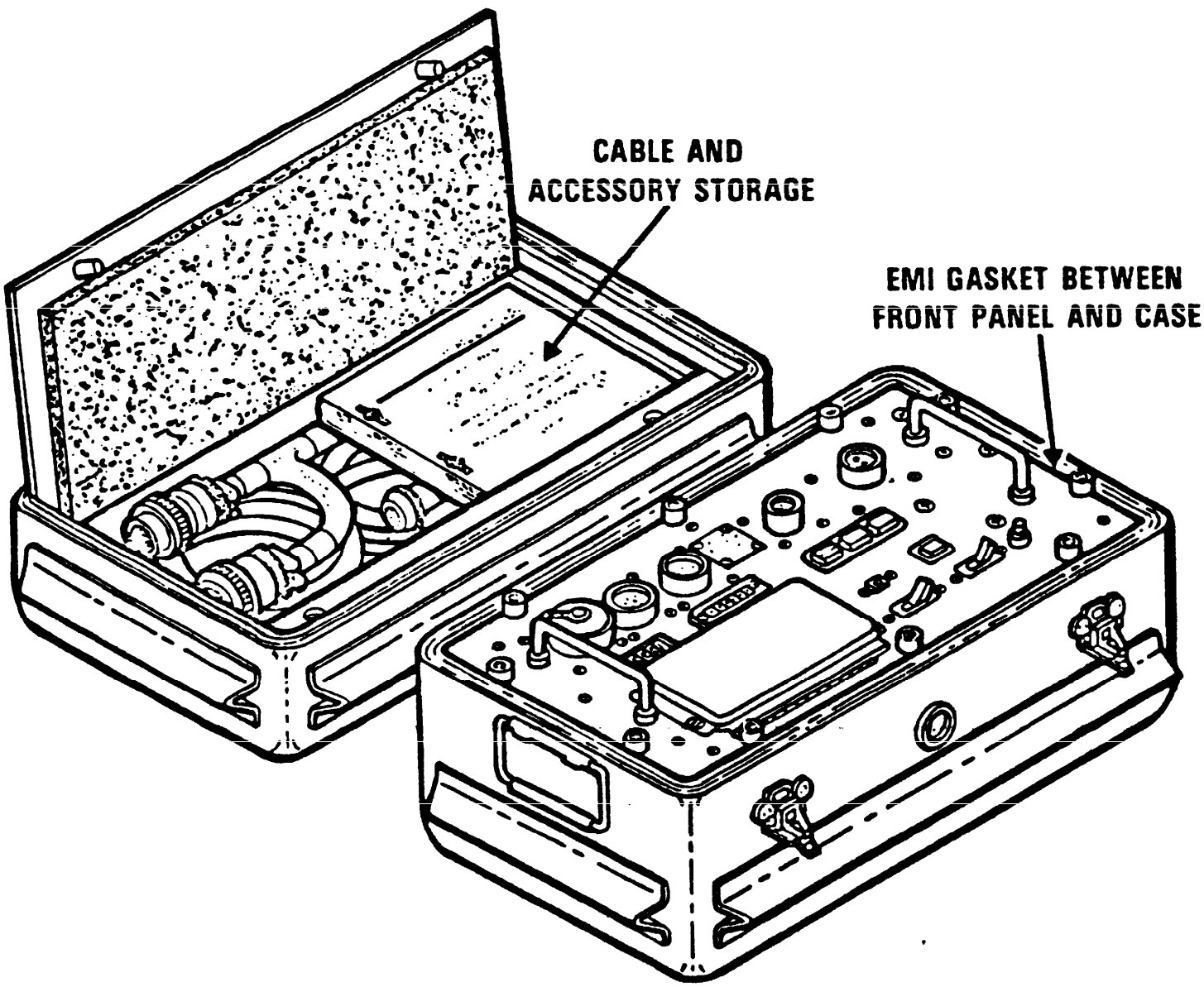


FIGURE 2. Test set, fire control subsystem

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3.2.5 Electrical power. The FCC-TTS shall receive power from a 28 (+1, -6) volt, 16A maximum external power source. The FCC-FTS internally converts the dc source voltage to +5v for the logic supply voltage and -15v and +15v for the 400Hz reference circuit. It also supplies 28 (+1, -6) volts to the FCC-FTS to be tested. Overload protection of the 28 Vdc shall be provided by circuit breakers.

3.2.6 Physical characteristics. The Test Set shall be a portable unit housed in a combination case to protect it from adverse environment. The unit shall be self-contained and provide storage for all cables and the instruction manual.

3.2.6.1 Weight. FCC-FTS weight shall be 54 pounds maximum.

3.3 Material process and parts.

3.3.1 Microelectronic devices. Microelectronic devices shall conform to MIL-M-38510.

3.3.1.1 Soldering. All soldering shall be done in accordance with MIL-S-45743 or MIL-S-46844.

3.4 Electromagnetic radiation. The Test Set shall conform to the BMI requirement of MIL-T-28800, Paragraph 3.9.12 for Class 4 equipment and MIL-STD-461.

3.5 Interchangeability. Interchangeability shall be in accordance with MIL-STD-454, Requirement 7, and MIL-T-28800, **Paragraph** 3.12.1.

3.6 Safety. Safety requirements shall be in accordance with MIL-STD-28800, Paragraph 3.2 for Typ 1 equipment.

3.7 First article inspection. This specification contains technical Provisions for first article inspection. Requirements for the submission of first article by the contractor shall be as specified in the contract.

3.8 Preconditioning (burn-in). Each FCC-FTS procured under this specification shall be subjected to a burn-in test as defined in Figure 3.

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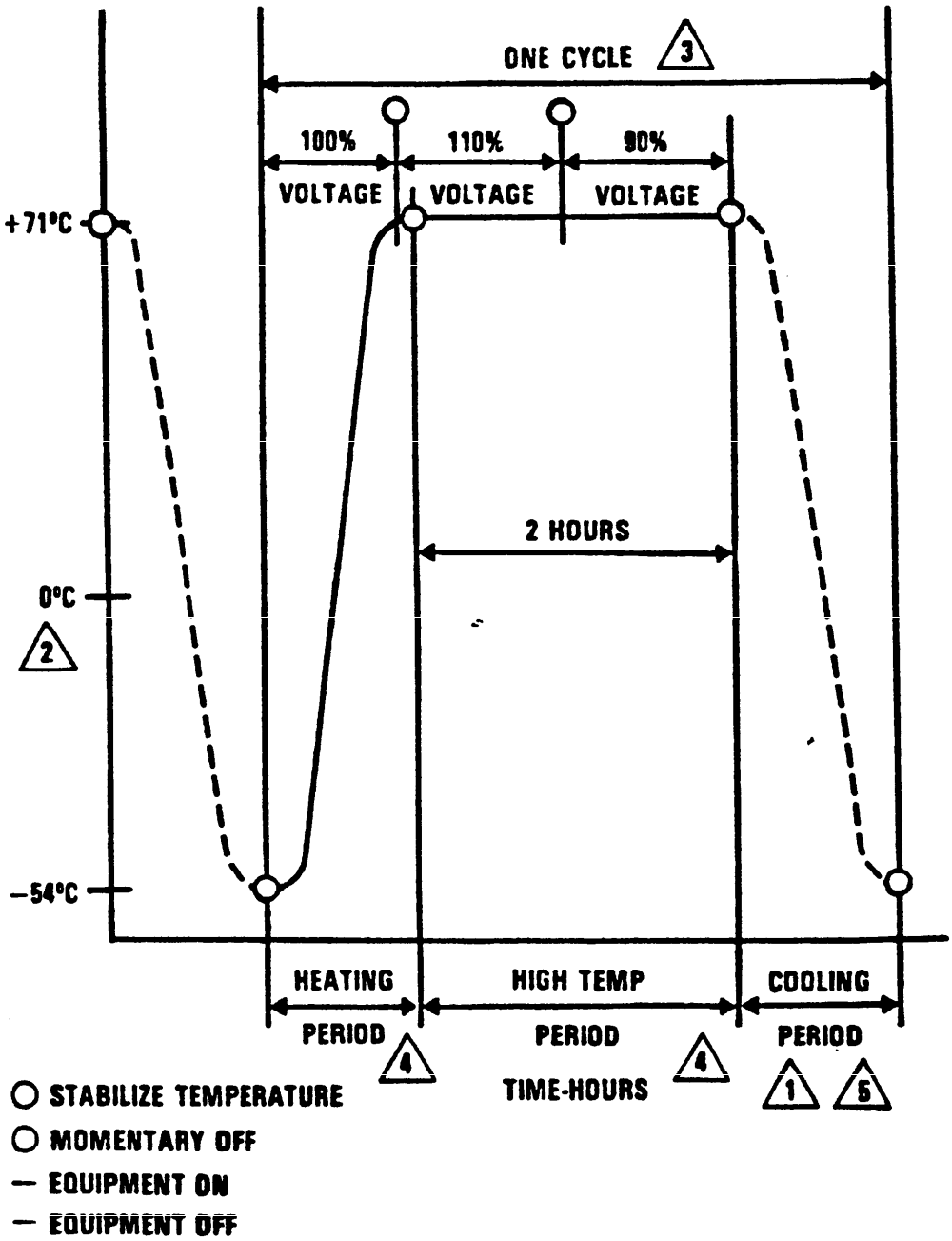


FIGURE 3. Typical temperature cycle - electronic and avionic equipment

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NOTES :

1. During the cooling period, the chamber temperature shall not be lower than -600C.

2. Once every 24 hours at OoC, moisture shall be **injected until the equipment being tested is coated with frost.**

3. One temperature cycle shall start with the equipment stabilized at the low temperature, heated and stabilized at the high temperature, maintained at the high temperature for 2 hours, then end with the equipment cooled and stabilized at the low temperature. During the temperature cycle, the equipment shall be operated in accordance with its normal duty cycle.

4. Equipment ON; input **voltage at 110% and 90%, respectively during each 1/3 of the total time of heating period plus high temperature period; momentarily** OFF **between input voltage changes.**

5. Equipment OFF.

6. Vibration -2.2g ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The Vibration shall be at least 10 minutes during each hour of operating time.

7. Transition rate between temperature levels shall average 5oC per minute.

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3.9 Environmental conditions (first article testing only). The **FCC-FTS shall operate and** meet the performance requirements of 3.2.1 under any of the environmental conditions, or a reasonable combination of these conditions as specified herein.

3.10 Temperature/altitude.

3.10.1 Altitude. Operating, 0-10,000 feet; nonoperating and storage, 0-10,000 feet.

3.10.2 Temperature. operating, 0 to +55°C; nonoperating and storage, -62° to +85°C.

3.10.3 Humidity. operation while being subjected to MIL-STD-810 Qs modified by MIL-T-28800, Method 507.1.

3.10.4 Vibration. The FCC-FTS unit shall not sustain damage or subsequently fail to provide specified performance when subjected to the vibration environment defined by MIL-STD-810, Method 514.2.

3.10.5 Shock. The **FCC-FTS unit** shall not sustain damage or subsequently fail to provide specified performance when subjected to shock as defined by MIL-STD-810 and MIL-T-28800, Method 516.2.

3.10.6 Handling. The FCC-FTS unit shall not sustain damage or subsequently fail to provide specified performance when subjected to a 4" drop to a **wood surface.**

3.10.7 Transit drop. The FCC-FTS unit shall not sustain damage or subsequently fail to prove specified performance when subjected to a drop from 12" for a total of 26 drops.

3.10.8 Watertight requirement. The FCC-FTS shall not sustain damage or subsequently fail to provide specified performance when subjected to water submersion at 1.3 psi for 1 hour-

3.10.9 Salt **atmosphere. Operation after being subjected to MIL-STD-810, Method 509.1. Operation of the FCC-FTS shall not be required during testing and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered or otherwise loosened finish** of metal surface.

3010.10 FUNGUS **No evidence** of **viable** fungus on any surfaces after being subjected to MIL-STD-810, Method 508.1.

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3.10.11 Explosive atmosphere. Operation after being subjected to MIL-STD-810, Method 511.l, procedure I, excePt test to be limited to a maximum altitude of 20,000 feet

3.10.12 Sand and dust. Operation after being subjected to MIL-STD-810, Method 510.1, Procedure I. Step 3 may proceed immediately after stabilization in Step 2$_8$ if desired.

3.10.13 Workmanship. The workmanship shall conform to the general requirements of MIL-T-28800 and MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Respo**nisibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. **Except as** otherwise specified in the contract the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved **by the** Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections **are** deemed necessary to assure supplies conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements **of** the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein **are** classed as follows:

a. **First article inspection (see 4.4).**

b. Quality **conformance inspection (see** 4.5).

4.3 **Inspe ction conditions. Unless otherwise specified** herein, all inspection shall be performed in accordance with the quality **assurance provisions of MIL-F-13926.**

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4:$ First article. The requirement for first article approval and the designation of responsibility for first **article** inspection **to either** the Government or the contractor shall be specified in the oontract.

4.4.1 Sample. The first article sample shall be three **FCC-FTS'S selected at random by the Government representative from the first fifteen production FCC-FTS'S.**

4.4.2 Inspection. **The sample shall be subjected** to all the inspections specified in Table I.

TAB LE 1. First article tests

I tern	**Characteristic**	Requirement	Test procedure
1	Performance	3.2.1	4.5.1, 4.6, 4.7
2	Soldering	3.3.1.1	MIL-S-45743 or MIL-S-48644
3	Electromagnetic Radiation	3.4	4.8.1.12, MIL-T-28800, and MIL-STD-461
4	Interchangeability	3.5	MIL-T-28800 & MIL-STD-454
5	Preconditioning	3.8	4.501, 4.6, 4.7, 4.8.1.14
6	Altitude	3.10.1	4.8.1.2, 4.5.1, 4.6, 4.7
7	Temperature	3.10.2	4.8.1.2, 4.5.1, 4.6~ 4.7
8	Humidity	3.10.3	4. 8.1. 3,5 4.5.1, 4.6, 4.7
9	Vibration	3.1004	4.8.1.4, 4.5.1, 4.6, 4.7
10	S hock	3.10.5	4.8.1.5, 4.501, 4.6, 4.7
11	Handling	3.10.6	4. 8.1.68 4.5.1, 4.6P 4.7
12	Transit Drop	3.10.7	4.8.1.8, 4.5.1, 4.6, 4.7
13	Watertight	3.10.8	4.8.1.7, 4.501, 4.6, 4.7
14	Salt atmosphere	3.10.9	4.8.1.9
15	**Fungus**	3.10.10	4.8.1.10
16	Explosive atmosphere	3.10011	4.8.1.11
17	Sand and dust	3* 10.12	4.8.1.13
18	Workmanship	3.10.13	MIL-T-28800 & MIL-STD-454

TABLE 11. Performance tests

I tern	Characteristic	Requirement	Test procedure
1	Performance	3.2.1	4.5.1, 4.6, 4.7
2	Soldering	3.3.1.2	MIL-S-45743 or MIL-S-46844
3	Interchangeability	3*5	MIL-T-28800 & MIL-STD-454
4	Preconditioning	3.8	4. 8.1.4, 45.1, 4.68 407
5	Vibration	3.10.4	4.8.1.4, 4.5.1, 4.6, 4.7
6	S **hock**	**3.10.5**	**4.0.1.5, 4.5,1, 4.6, 4.7**
7	**Transit Drop**	**3.10.6**	**4.8.8, 4.5.1, 4.6, 4.7**
8	**workmanship**	**3.10.13**	**MIL-T-28800 & MIL-STD-454**

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4.5 **quality** conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each FCC-FTS in every lot shall be subjected to the calibration procedure (Appendix A), and to the tests in Table II and shall be examined visually for completeness, improper assembly, and evidence of poor workmanship.

4.5.1.2 Acceptance. Where any FCC-FTS fails to meet any specified requirement the defective FCC-FTS shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.1.3 Test equipment. The test equipment listed in Table III will be required to complete the functional performance and environmental tests specified in this test procedure. The equipment may be changed, at the discretion of the coqnizant test engineer, with equivalent or superior rated equipment when the specified equipment is not readily available for the performance of the test.

The actual test equipment used during the performance of the test shall be recorded on the applicable data sheets. The nomenclature, type, **model, and identification numbers shall be recorded for each equipment used.**

All measurements shall be made with instruments and test equipment whose accuracy and calibration has been verified periodically to meet MIL-STD-45662 requirements.

All instruments and test equipment shall be used in accordance with the manufacturers approved procedures and recommendations.

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TABLE III. Comprehensive list of test equipment

Item	Description	Range/Accuracy	Type/Model	Alternate
1.	Voltmeter, TRMS			Equivalent
2.	Cobra FCC		TSC Design	None
3.	Temperature/ Altitude Chamber and Recording System	-80 to +200°C/ + 2°C, -lK to +10°K ft/ ± 5%	Hieatt/TW-27	Equivalent
4.	Random-Sine Vibration and Data Acquisition System	5 to 3 KHz, 25 lbs to 150 g, 70 in./ sec/, 1 in. da.	Unbolts-Dickie TA-16400-150	Equivalent
5.	Mechanical Shock and Data Acquisition System	O to 750 g. 100 lbs "	LAB/SD-16-42-100	Equivalent
6.	Humidity Chamber and Recorder System	-100 to +275°F/ ± 2°F RH to 100%	Conrad/FD-8-3-3	Equivalent
7*	Stop Watch	0-60 Sec/mins/~ 0.1%	Minerva	Equivalent
8.	Scope Camera	N/A	Polaroid F\l.9	Equivalent
9.	Multi-point Temperature Recorder	-100 to +2000F ± 1.2°F	Honeywell Electronic 16	Equivalent
10.	oscilloscope	o-5 MHz ± 3%	Tektronic	Equivalent

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TABLE III. Comprehensive list of test equipment, Continued

Item	Description	Range/Accuracy	Type/Model	Alternate
11.	Frequency Counter	HP 5245	HP	Equivalent
12.	Eletrical power	115 VAC, 60 Hz 10, 30A	Facility	Equivalent
13.	Electrical Power	115 VAC, 400 Hz 10, 30A	Facility	Equivalent
14•	Quick Release Hook	N/A	Facility	Equivalent
15*	Power Supply, DC	O-36 VDCt 5 AMP	HP/6266A	Equivalent
16.	Water Tank	3 feet dia. s feet deep	AETL Mfg .	Equivalent
17•	Ruler	36 inches	N/A	Equivalent

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4.6 Performance tests. Each Test Set submitted..for acceptance shall be subjected to the following tests to determine the functional capabilities. All tests shall be performed with the W2 and w5 cable assemblies connected and measurements made at the designated Test Set TEST POINTS or via the TEST RESULTS display as herein specified. Test Set power supply has been verified through the performance of calibration procedure Appendix A.

4.6.1 Cable assembly continuity tests. Continuity in the five cable assemblies furnished with each FCC-FTS shall be verified per the applicable wiring document.

4.6.2 FCC-FTS test points. Verify electrical continuity between points listed in Table IV using DVM set to K ohms. Record results in data sheets.

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TABLE IV. FCCY,FTS test 20 .

E&E	Test Point*	Corm. Pin	Name		Test Point	Corm. Pin	Name
1	TP-AOl	S1-1	EXTM*	29	TP-C05	J1-90	10 KHz
2	TP -A02	J1-3	MEND*	30	TP-C06	J1-96	RINH
3	TP-A03	J1-5	TCLI)	31	TP-C07	J1-98	R INT
4	TP-AO 4	J1-7	TECK *	32	'1'P-C08	J1-100	HALT*
5	TP-A05	S1-30	TPHA	33	TP-C09	J1-100	A3AA
6	TP -A06	J1-32	TPHB	34	TP-C10	J1-102	PH3B
7	TP-A07	J1-34	TPHC	36	TP-cll	J1-108	TIBA*
8	TP -A08	J2-22	HUD	36	TP-C12	J1-110	TIBB*
9	TP-A09	J2-23	HUD	37	TP-!)01	J1-112	TIBC*
10	TP-A10	J2-5	DACK	38	TP-D02	J1-114	+ 5VM0
11	TP-A1l	J2-6	DACK*	39	TP-D03	J1-92	+15VDC
12	TP-A12	J2-9	EASM	40	TP-D04	J1-93	COM/GRD
13	TP-BO1	J2-10	EASM*	41	TP-D05	J1-94	-15VDC
4	TP-B02	J2-3	DXMRQ	42	TP-D07	J1-117	DCMUXR
15	TP-B03	J2-4	DXMR()*	43	TP-D08	J1-118	DCMU X
16	TP-B04	J2-14	DM?llY	44	TP-D10	J5-90	ST +15
17	TP-B05	J2-15	DliRDY*	45	TP-D1l	J5-93	ST GRD
18	TP-B06	J2-1	DAT'A	46	TP-D12	J5-94	ST -15
19	TP-B07	J2-2	DATA*	47	TP-EO1	J2-20	ACREF
20	TP-B08	J2-16	LRF	48	TP-E02	J2-21	ACREF
21	TP-B09	J2-17	LRF*	49	TP"E03	J3-A	+ 28vDC
22	TP-B10	J2-18	DOPP	50	TP-E04 '	J3-c/D	+28VR
23	TP-B1l	J2-19	DOPP*	51	TP-E05	J1-126	Muxl
24	TP-B12	071-60	DICA*	52	TP-E06	J1-127	MUX2
25	TP-col	J1-61	DICB	53	TP-E09	J1-97/99	GRD
26	TP-C02	J1-63	DICC*	54	TP-E10	J1-101/ 103/105	GRD
27	TP-C03	J1-64	DICD*				
28	TP-C04	J1-66	DIcE*	55	TP-XZI1	J1-109/ 111/113	GRD

Connect W5 Cable to Jl, J2, and J5 of the Test Set as indicated on the cable.

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4.6.3 Power on tests.

4.6.3.1 Power supply test. Turn on the 28 VDC, +15 VIZ, and -15 WC power supplies. Depress the POWER switch and observe the following:

POWER switch indicator is ON.
Time totalizing meter is running.

Record the results in data sheet.

4.6.3.2 Test Point voltages. Using DVM set to AUTO DC Mode, verify voltages to Table V. Record results in data sheets.

TABLE V. Test point voltages

HI	RTN	Specified Limits
TP-D03	TP-D04	+13.5 to +16.5 VDC
TP-D05	**TP-D04**	**-13.5 to -16.5 VDC**
TP-E03	**TP-E04**	**+22.0 to +30.8 VDC**
TP-E07	TP-E08	+4.5 to +5.5 WC

4.6.3.3 10 **KHz** clock. Using scope, measure 10 **KHz + 10% clock at TP-C05 (HI) (STIOK*) and TP-E12 (GRD). Record result-in data sheets.**

4.6.4 Power switch test. Depress the POWER switch and observe the following.

POWER switch indicator is OFF.
Time Totalizing meter is stopped.

Record the results in data sheet.

4.6.5 **Lamp test. Depress POWER switch (indicatro should now be on and should remain on for all of the subsequent Powered Tests)- Depress the LAW TEST switch and observe the following.**

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MODE - START indicator: ON

MODE - SINGLE indicator: ON

MODE - LOOP indicator: OH

TEST RESULTS display: All segments on (888888)

Record the results in the data sheet.

4.6.6 Memory test. Set the TEST SELECT to 000; depress the MODE-START switch and observe the following reading on the TEST RESULTS display.

010037

Record the results in data sheet

4.6.7 TEST SELECT and MODE tests.

4.6.7.1 Test 100. Set the TEST SELECT thumbwheel switches to 100 and observe the following reading on the TEST RESULTS display:

002100

Note: If the TEST RESULTS display reading is 101106, 001100, or 102100, alternately depress the MODE-SINGLE/LOOP and/or START switched until the display reading is 002100.

Record the results in data sheet.

4.6.7.2 Test 117. Set the **TEST SELECT thumbwheel switches to 117** and observe the following reading on the TEST RESULTS display.

002117

Record results in data sheet.

4.6.7.3 Test 126. Set the TET SELECT thumbwheel switches to 126 and observe the following reading on the TEST RESULTS display.

002126

Record **the** results **in the data sheet.**

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4.6.7.4. Test 135. Set the TEST SELECT thumbwheel switches to 135 and observe the following reading on the TEST RESULTS display

002135

Record the results in the data sheet.

4.6.7.5 Test 144. Set the TEST SELECT thumbwheel switches to 144 and observe the following reading on the TEST RESULTS display.

002144

Record the results in data sheet.

4.6.7.6 Test 153. Set the TEST SELECT thumbwheel switches to 153 and observe the following reading on the TEST RESULTS display.

002153

Record the results in data sheet.

4.6.7.7 Test 162. Set the TEST SELECT thumbwheel switches to 162 and observe the following reading on the TEST RESULTS display.

002162

Record the results in data sheet.

4.6.7.8 Test 171. Set the TEST SELECT thumbwheel switches to 171 and observe the following reading on the TEST RESULTS display.

002171

Record the results in **data sheet.**

4.6.7.9 SINGLE\LOOP test. **Set the TEST SELECT thumbwheel switches to 171 (as in paragraph 4.6.7.8) depress the MODE-SINGLE/LOOP switch; and observe the following reading on the TEST RESULTS display.**

101171

Record the results in data sheet.

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4.6.7.10 *MODE/START* test. The following action sequence -is predicated upon successful completion of test per paragraph 4.6.7.9 and required utilizing the same TEST SELECT setting as specified herein.

a. Depress MODE-START switch, then release it.

b. **Depress MODE-START switch again and hold depressed while observing the following reading** on the test results displays then **release it.**

105171

Record the results in data sheet.

4.6.8 Operational Test analysis (OTA) 71 test.

4.6.8.1 Test 207. Set the **TEST SELECT thumbwheel switches to 207 and observe all MODE indicators are off.**

Record the results in data sheet..

4.6.8.2 Test **200. Set the TEST SELECT thumbwheel switches to 200 and observe the following MODE indications:**

START : Extinguished
SINGLE : Illuminated
LOOP : Extinguished

Record the results in data sheet.

4.6.8.3 Test 201. Set the TEST SELECT thumbwheel switches to 201 and observe the following MODE indications:

START : Extinguished
SINGLE : Extinguished
LOOP : Illuminated

Record the results in data sheet.

4.6.8.4 **Test 202. Set the TEST SELECT thumbwheel switches to 202 and observe the following** MODE indications:

START : Illuminated
SINGLE : Extinguished
LOOP : Extinguished

Record the results in data sheet.

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4.6. 8.5 Test 203. Set the TEST SELECT thumbwheel switches to 203 and observe the following reading on the TEST RESULTS display.

101171

Record the results in data sheet.

4.6. 8.6 Test 103. Set the TEST **SELECT** thumbwheel switches to 103 and observe the following reading on the TEST RESULTS display.

131103 .

Record the results in data sheet.

4.6.9 Operational test analysis (OTA) 72 test.

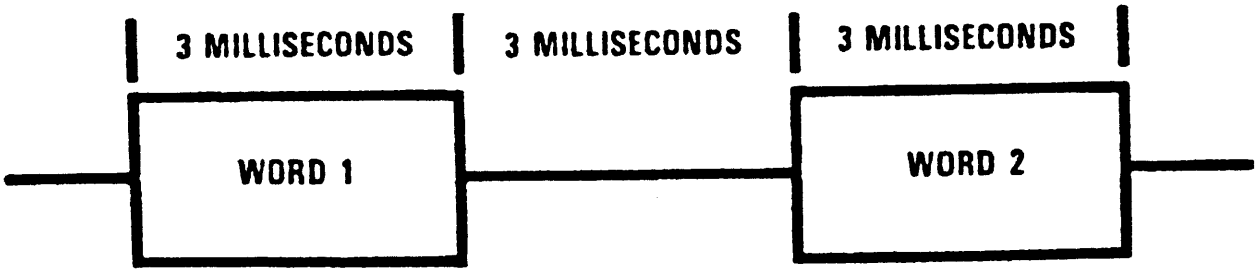
4.6.9.1 Test 300. Set **the TEST SELECT thumbwheel switches to 300. Utilizing a suitable oscilloscope (set-uP as specified below and connected to the stipulated TEST POINTS, observe the waveforms per Figure 4.**

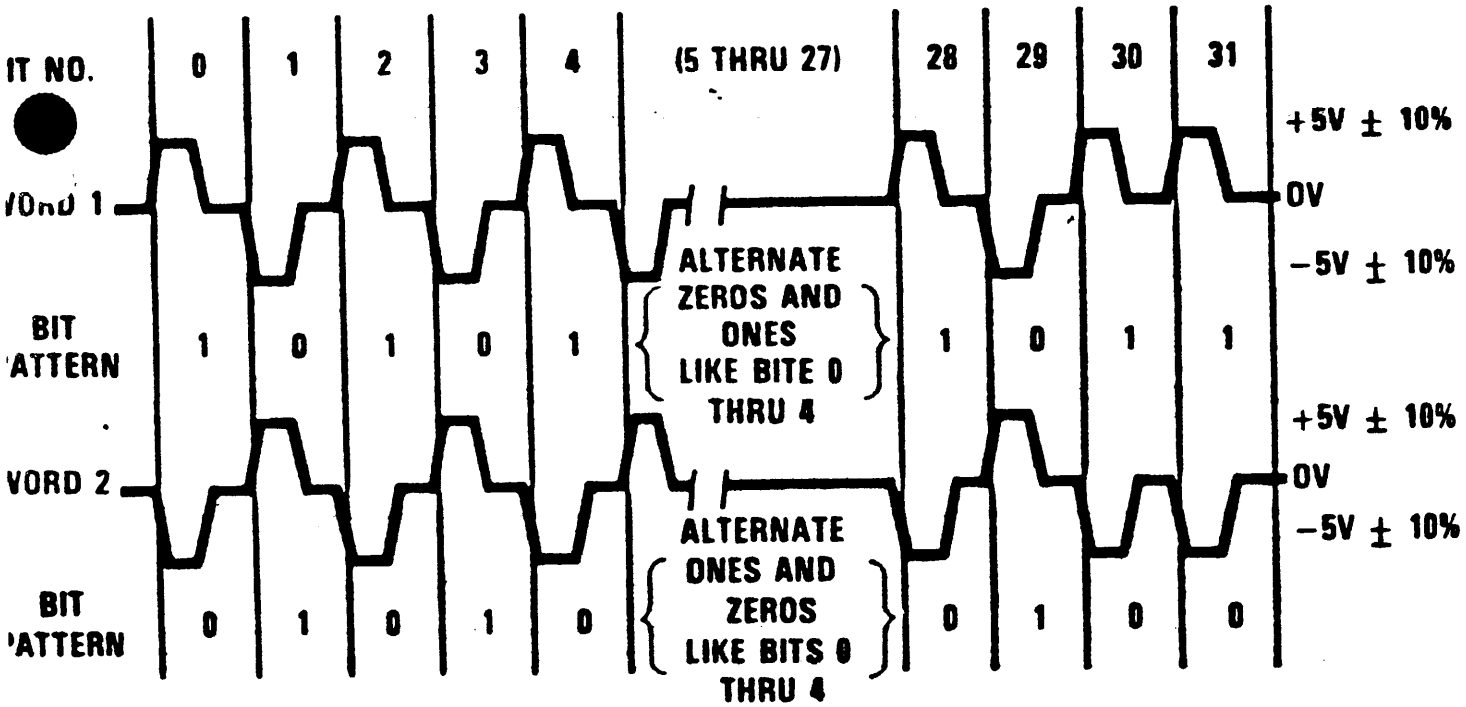
Oscilloscope **Channel 1**	**Test Set** TP-B10 **DOPP**
Oscilloscope Trigger Mode	**AUTO**
Oscilloscope Mode **Display**	**A**
Oscilloscope **Trigger Source**	**Channel 1**
Oscilloscope Coupling	**AC**
Oscilloscope Time Base	**1 millisec/div.**
Oscilloscope Channel 1 1	DC 5v/piv
Oscilloscope Vert. Mode"	**Channel 1**

Record the results in data sheet. Repeat again with oscilloscope channel 1 on TP-B1l. The waveform will be the complement Of -TP-I310. Record the results, in data sheet.

4.6.9.2 **TP-B10 Tests' Change the oscilloscope Time Base setting, per paragraph 4.6.9.1, to x10 Mag (in) and observe that the waveform at TP-B10 are "as described in Figure 4. Record the results in data sheet.**

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TIME FROM BIT 0 TO BIT 31 = 3 MILLISEC ± 10%

TIME FROM BIT 31 WORD 1 TO BIT 0 WORD 2 = 3 MILLISEC ± 10%

FIGURE 4. Bit word

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4.6.10 Operational test analysis (OTA) 73 test.

4.6.10.1 Data Pattern tests Set the TEST SELECT thumbwheel switches as specified below. Utilizing the oscilloscope used in test per paragraph 4.6.9 (set-up as specified therein, except as changed in this paragraph) and connected to the stipulated TEST POINTS, observe the required waveforms.

Oscilloscope Channel 1	DC 2V\Div
Oscilloscope Channel 2	DC 2V/Div
Oscilloscope Time Base	As required
Oscilloscope Vert. Mode	ALT

NOTE : Output signals of Test Set will all be formal T^2 levels (i.e., Logic "0" = 0 to 0.4V - Logic "1" = 2.4 to 5.0V).

a. TEST SELECT Setting: 402

Signal	scope Chan.	Test Set TP	Required Data Pattern
DARDY	1	B04	=0.5 sec puls every 40 millisec ± 10%
DARDY	2	DOS	Complement of DARDY

Record the results in data sheet.

b. TEST SELECT setting: 492

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DATA	1	B06	Table VI
DATA*	1	D07	Complement of DATA

Record the results in data sheet.

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TABLE VI. Data Pattern

WORD NO.	1st	2nd	3rd	4th	5th	6th	Parity (49th Bit)
PATTERN	01010101	10101010	01010101	10101010	01010101	10101010	1

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UIEQ	Scope Chan.	Test Set TP	Required Data Pattern
LRF	1	B08	Table VI
LF2F	2	D09	Complement of LRF

Record the results in data sheet.

d. TEST SELECT Setting: 401

Signal	Scope Chan.	Test Set TP	Required Data Pattern
EASM	1	A12	**Table VI**
EAS MA	2	BO1	**Complement of EASM**

Record the results in data sheet.

e. TEST SELECT Setting: 400

Signal	Scope Chan.	Test Set TP	Required Data Pattern
LRF	1	B08	**20 microsec logic** "O" , every **100 microsec** ± 1 0%
LRF	2	B09	**Complement of LRF**

Record the results in data sheet.

4.6.11 Memory control test.

4.6.11.1 Waveform test. Set the TEST SELECT thumbwheel switches as specified below. Utilizing the oscilloscope,us~ in tests per paragraph 4.5.1.4.7 (set-up as **specified therein except as changed in this paragraph) and connected to the stipulated TEST POINTS, observe the required waveforms.**

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a. **TEST SELECT setting: 000**

Signal	Scope Chan.	Test Set TP	Required Data Pattern
EXTM*	l or 2	A0 1	1 microsec pulse every 150 microsec ± 10%

Note: Logi c "1" **Superimposed on signal periodically.**

Record the results in data sheet.

b. **TEST SELECT Setting: 300**

Signal	Scope Chan.	Test Set Tp	Required Data Pattern
EXTM*	**l o r 2**	**A0 1**	2.4 VDC 5.0 VDC

Record the results in data sheet.

c. **TEST SELECT Setting: 300**

Signal	Scope Chan.	Test Set Tp	Required Data Pattern
DICA*	l or 2	B12	**2.4 VDC 5 VDC**

Record the results in data sheet.

d. **TEST SELECT Setting: 300**

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DICB*	**l o r 2**	**C01**	0.4 VDC

Record the results in data sheet.

e. TEST SELECT Setting: 300

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DICC*	l or 2	C02	0.4 VDC

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Record the results in data sheet.

f* TEST SELECT setting: 300

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DICD*	l or 2	C03	0.4 VDC

Record the results in data sheet.

90 **TEST SELECT Setting: 300**

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DICE*	l or 2	C04	2.4 VDC 5.0 VDC

Record the results in data sheet.

h. TEST SELECT Setting: 500

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DICA*	1012	B12	0.4 VDC

Record the results in data sheet.

i. **TEST SELECT Setting: 500**

Signal	Scope Chan.	Test Set Tp	Required Data Pattern
DICB*	l or 2	C0 1	2.4 VDC 5.0 VDC

Record the results in data sheet.

J **TEST SELECT Setting: 500**

Signal	Scope Chan..	Test Set TP	Required Data Pattern
DICC*	**l o r 2**	**C02**	2.4 VDC 5.0 VDC

Record the results in data sheet.

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k. TEST SELECT Setting: 500

Signal	scope Chan.	Test Set TP	Required Data Pattern
DICD*	1 or 2	C03	2.4 VDC 5.0 VDC

Record the results in data sheet.

1. **TEST SELECT Setting: 500**

Signal	Scope Chan.	Test Set TP	Required Data Pattern
DICE*	1 or 2	C04	0.4 VDC

Record the results in data sheet.

4.6.12 400 Hz reference signal test.

4.6.12.1 400 Hz test (oscilloscope). Set the oscilloscope time **base to 1** mSec/Div; connect Channel 1 to Test Set TP-EOl (AC Refernece); Ground Reference to **ANALCXS GND (AC Reference RTN), Test Set TP-EOZ; and observe a 400 Hz, 28V pp ± 5% sinewave. Record the results in data sheet.**

4.6.12.2 400 Hz Test (counter). Connect the frequency counter to TP-EC1-(HI), and TP-E02-(LO) and observe that the frequency is 400 Hz ± 10% Record the results on the data sheet.

4.6.13 End memory control test. Depress the POWER switch on the Test Set Panel. Turn off -15 VDC, +15 VDC, and 28 **VDC pwer** supplies. Remove all ?.cads from Test Set TEST POINTS and close and secure the TEST POINT cover . Disconnect the Cable Assemblies, and return them to their stowage locations in lid of Combination Case. Remove the ground connection from Test Set panel to Power Supply Ground. Replace cap on J5. Reassemble **lid and lower portion of** Combination Case. Close and **secure combination case.**

4.7 Self test. Utilize the **FCC-FTS "Self-Test" test** set-up shown in Figure 5 and the FCC-FTS face panel layout and nomenclature of Figure 1 to perform the **following test.** With the "Power" switch lamp "ON", set the "Test Select" thumbwheel$ to "000" now depress the "Mode" group "Start" switch. When **the "Test** Results" display indication stabilizes, the number should indicate 010037. **Record** the numbers **from each segment on the Data Sheet. This "Test Result" display indication will be repeated at the end of each 'Self-Test" cycle.**

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4.8 Performance test under simulated environmental conditions.

4. 8.1 Environmental test requirements and Procedures.

4. 8.1.1 Test sequence. The FCC-FTS shall have completed the test of paragraph 4. 5.1 prior to performance of the detailed environmental tests herein. The performance test of paragraph 4. 5.1.5 shall be performed before, during. or after each environmental test as required for the purpose of determining the performance of the FCC-FTS. The test of paragraph 4. 5.1 shall again be performed after all environmental tests have been completed

4. 8.1.2 Ternperature/altitude. The FCC-FTS shall be subjected to the applicable tests required by MIL-STD-810 with the temperature requirements modified by MIL-T-28800. The applicable tests are listed in Table VII with cross-reference to MIL-STD-810 and test paragraphs in this procedure. Table VII is a summary of the Tmp/Alt conditions which the equipment will experience. These conditions relate directly to MIL-STD-810, Method 504.1, Preocedure 1, for Category 2 equipment. Category 2 results from the MIL-T-28800 paragraph 4.5.5.2 requirement for operation of 10,000 feet maximum. Temperature limits-have been modified per MIL-T-28800, Table XIII, for Class 2 equipments.

4.8.1.2.1 Test facility.

a. Tem perature/altitude chamber. The temperture/altitude chamber shall be capable of maintaining the specific ambient temperature and altitude.

b. Thermocouples. The FCC-FTS shall be instrumented with thermocouples located on all components within the unit which are expected to exhibit the greatest thermal inertia during the test. **Paragraph 4.8.l.2.3(a).**

c. **Multi-poin t recorder. A multi-point~ strip chart recording system, capable of monitoring and recording the temperature changes** sensed by the thermocouples shall be utilized for plotting the **thermal changes versus time for each temperature critical component.**

4. 8.1.2.2 **Tempe rature/altitude test equi Pment. The equipment listed in Table VIII will be required to conduct the teraperature/ altitude tests. All measurements shall be made with instruments whose accuracy and calibration has been verified periodically and meet specification requirements.**

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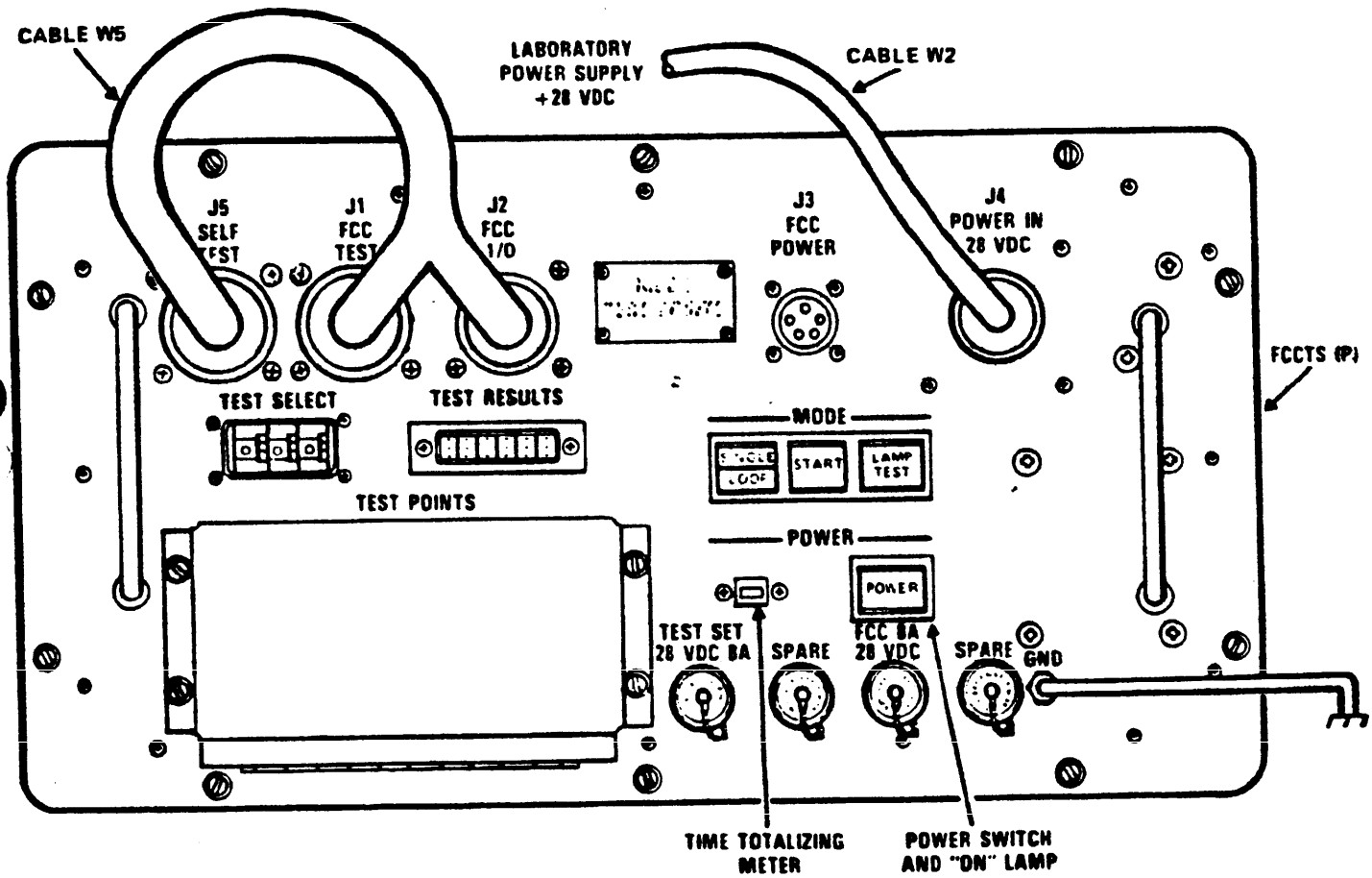


FIGURE 5. FCCTS (P) system performance test - test set-up

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TABLE VII. Temp/Alt test condition summary

Step Per MIL-STD-81 o Proc I Cat 2		Alt	Ope r	Test Step
l(b)	-62	Site	No	a
2	0	Site	Yes	b
3	0	1000	Yes	c
4	0	Site	Yes	d
5	*standard	Ambient	Yes	e
G	85	Site	No	f
.	55	Site	Yes	9
8	55	Site	Yes	h
9	52	10000	Yes	i
10	55	10000	Yes	;
11	Standard	Ambient	Yes	

*standard ambient conditions are:

Temperature 230 ± 10°C

Relative Humidity 50% +30%

Atmosphere Pressure 75 $^{+50}_{-75}$ mm Hg

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4. 8.1.2.3 Test setup.

a. The FCC-FTS shall be instrumented with small diameter thermocouples placed on components within the unit, which are expected to exhibit the greatest thermal inertia during the test. All components instrumented with thermocouples shall be identified and listed on the applicable data sheet. The thermocouple leads shall be routed to the outside of the fCC-F'lS by any convenient means.

b. The test setup and methods of operation of the FCC-FTS shall be such as to simulate actual usage insofar as practical. The fCC-FTS undergoing tests shall be placed in the test chamber with electrical and thermoscope leads made through altitude sealed terminals, connectors, and parts. The thermocouple leads shall be connected to a multi-point temperature recorder.

Notes: The following guidelines are provided for consideration when determining location of thermal sensors used to monitor the test items:

(1) One or more sensor in the ambient air within each major unit.

(2) Contact temperature on the largest mass in each major unit.

(3) Contact temperature on the part(s) where the highest surface temperature is expected.

(4) Contact temperature on the part(s) whose temperature is likely to limit equipment performance.

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TABLE VIII. Temperatur e,/altitude test equipment

Description	Range/Accuracy	Type/Model	Cal ibr at i on period
Temperature/Altitude Chamber and Recording	-80 to +200°/~2°C - lK to +100K feet/~5%	Hieatt/THV-27	As Req'd
S top Natch	0-50 sees/mins/+0. 1%	Minerva	As Req'd
Multi-point Temperature e Recorder	-1oo to +200°F +~1 . 20F	Honeywell/ Ekectroni k 16	As Req'd
Oscilloscope	0-5 MHz ± 3%	453	As Req'd
Vertical Amp Plug in	DC -24MHz 23%	CA	As Req'd
Electrical Power	115 vac, 60 Hz 10, 30A	Facility	N/A
Power Supply, DC	0-36 volts/5amps	HP/6266A	As Req'd

4. 8.1.2.4 Temperature/altitude test methods of inspection.

4. 8.1.2.4.1 Test performance. The chamber air will be controlled to the specified ambient temperatures with all temperatures monitored and recorded. With the FCC-FTS placed in the temperature/ altitude chamber and connected as specified in paragraph 4.5.2.1.5? determine initial conditions as outlined in 4.5.2.1.6.2.

4. 8.1.2.4.2 Standard conditions (reference) . With the chamber door open and the FCC-FTS exposed to room ambient conditions, the FCC-FTS shall be turned on and operated in accordance with the functional tests of paragraph 4.7. The functional performance of the FCC-FTS shall be checked with warmup times not exceeding 10 minutes and the input voltages to the FCC-FTS set at:

a. Nominal 28 VDC

b. High 30.8 VDC

c. Low 25.2 VDC

The multi-point temperature recorder shall be on and recording the chamber wall and air temperatures and the thermally instrumented component temperatures wi thin the FCC-FTS throughout all temperature/ altitude tests. Record the results of the above tests on the applicable data sheets.

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4. 8.1.2.4.3 Environmental tests. Subject the FCC-FTS to the tests outlined in the following steps:

Step a. -620 AtmoSpheriC

With the test item de-energized, adjust the chamber test conditions to -62°C and ambient preSSure. After the test item temperature has stabilized, maintain these test conditions for at lease two hours. Where it is possible, without changing the test conditions. the test item shall be visually inspected for the presence of deterioration which would impair future operation.

Step b. 0° Atmospheric

With the test item de-energized, adjust the chamber test conditions to OoC and ambient pressure. These chamber test conditions shall be maintained throughout this step. After the test item temperature has stabilized, the test item shall operate satisfactorily immediately following the specified warmup time.

All characteristics which are likely to be affected by low temperatures shall be checked first. Should the time required to check the test item exceed 15 minutes beyond the warmup time, the test item shall again be stabilized at .

The test item shall then be de-energized and re-established at the temperature specified. The above operational sequence shall be repeated two additional times.

Step c. OoC, 10,000 Feet

With the test item de-energized, adjust the chamber temperature to OoC. The test item temperature shall be stabilized. The test shall be energized at the highest specified input voltage and the chamber pressure adjusted to simulate 10,000 foot altitude. Upon reaching the specified chamber pressure and while maintaining the chamber temperature, the test item shall be checked for satisfactory operation and the results recorded.

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Step d. 0°C Atmospheric

With the test item de-energized, adjust the chamber test conditions to 0°C and ambient pressure. After the test item temperature has stabilized, the chamber door shall be opened and frost permitted to form on the test item.

When the chamber door is opened, it is intended that frost will form; however, should the relative humidity of the air be such that frost will not form, an artificial means shall be used to provide the relative humidity necessary to have frost form.

The door shall remain open long enough for the frost to melt, but not long enough for the moisture to evaporate. The chamber door shall be closed.and the test item operated at the highest specified input voltage to ascertain satisfactory operation immediately following the specified warmup time. The test item shall be energized and de-energized at least three times.

Step e. Standard Ambient, Atmospheric

Adjust the chamber test conditions to standard ambient conditions After the test item temperature has stabilized, an operational and performance check of the test item shall be made and the results compared with the data previously taken and evaluated using the failure criteria of Sections 4.4 and 4.4.1.

Step f. 85°C Atmospheric

With the test item de-energized, adjust the chamber test conditions to 85°C and ambient pressure. The test item temperature shall be stabilized and then maintained for at least 16 hours. At the conclusion of this time period, where practicable, the test item shall be visually inspected for any deterioration.

Step g. 550 Atmospheric

With the test item de-energized, adjust the chamber test conditions to 55°C and ambient pressure. After the test item temperature has stabilized and while maintaining the chamber temperature, operate the test item continuously at

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the highest specified input voltage. Thermal sensor readings of the test item temperature shall be recorded every 30 minutes. At the end of four hours, and while maintaining the test conditions the test item shall be checked for satisfactory operation and the results recorded.

Step h. 55°, atmospheric

With the test item de-energized, adjust the chamber test conditions to 550C and ambient pressure, and maintain until copletion of the fourth operating time period. After the test item temperature has stabilized, the test item shall be operated at the highest specified input voltage for four time periods, each of 20 minutes. The first three time periods of operation shall be followed by a 15 minute period with the test items de-energized. The test item shall be checked for satisfactory operation during each operating time period and the results recorded. Thermal sensor readings of the test item temperature shall be recorded every 10 minutes of test item operation.

Step i. 52°, Atmospheric

with the test item de-energized, adjust the chamber temperature to 52°C. The test item temperature shall be stabilized. The test item shall be operated at the highest specified input voltage and the chamber pressure adjusted to simulate an altitude of 10,000 feet. Maintain these test conditions for four hours. Thermal sensor readings of the **test item** temperature shall be recorded every 30 minutes. At the end of the four hour period and while maintaining the test conditions, the test item shall be checked for satisfactory operation and the results recorded.

Step j. 55°C, 10,000 Feet

With the test item de-energized, adjust the chamber temperature to 55°C, and maintain until completion of the fourth operating time period. The test item temperature shall be stabilized. The test item shall be energized at the highest specified input voltage and the chamber pressure adjusted to simulate an altitude of 10,000 feet. (Following **those steps where a change in temperature at low pressure is required, the pressure may be increased to ambient before changing the temperature and then returned**

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to the required pressure following temperature stabilization.) The test item shall be operated for four time periods each of 20 minutes duration. The first three time periods of operation shall be followed by a 15 minute period with the test item de-energized. The test item shall be checked **for** satisfactory **operation during each operating time period and the results recorded. Thermal sensor readings of the test item temperature shall be recorded every 10 minutes of test item operation.**

Step k. Standard Ambient Conditions

Adjust the chamber test conditions to standard ambient conditions. After the test item temperature has stabilized, an operational and performance check **of the** test item shall be made and the results of the test on the applicable data sheet.

4.8.1.3 Humidity. The FCC-FTS shall be subjected to the applicable tests required by MIL-STD-810 as modified by MIL-T-28800. The test procedure is listed below and in Table IX and Figure 6.

4.8.1.3.1 Test facility.

a. Humidity chamber. The humidity tests shall be performed in a chamber capable of generating and controlling relative humidity in excess of 95% RH by steam or evaporation of tap water having pH value between 6.0 and 7.2 measure at 23°C. **The chamber shall also contain provisions for limiting** *air* **velocity throughout the exposure are to 150 feet per minute and provisions made to prevent dripping and collecting water-condensate onto the equipment fomn overhead. The chamber shall have been designed and constructed with provisions** for, 1) venting to atmosphere to preclude vapor pressure buildup and 2) sufficient internal volume so that the FCC-FTS **under test will not interfere with the generation and control of the test conditions.**

b. Humidity sensor. The relative humidity within the test chamber **shall be verified by use of one or more of the following type indicators:**

(1) Dew Point indicator (Dry bulb temperature and water content) .

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TABIE IX. Humidity tests

Test Conditions	MIL-STD-810 , Paragraph
1. Room Ambient Uncontrolled Humidity (non-operating)	Method 507 Procedure IV, 3.4
2. +450 + 50C 2 Hours (Min) (non-operating)	3.4
3. +25° + 5°C, RH 50% 24 Hours (non-operating)	3.4
4. 5 cycles, 24 Hours each 60° ± 5°C and 300 ± 5% 95% RH	3.4
5. +25° ± 5°C, 50% RH 12 to 24 Hours	3.4
6. Room Ambient Uncontrolled Humidity (operating)	3.4

2. Differential Temperature Indicator (Dry bulb temperature -) wet bulb temperature).

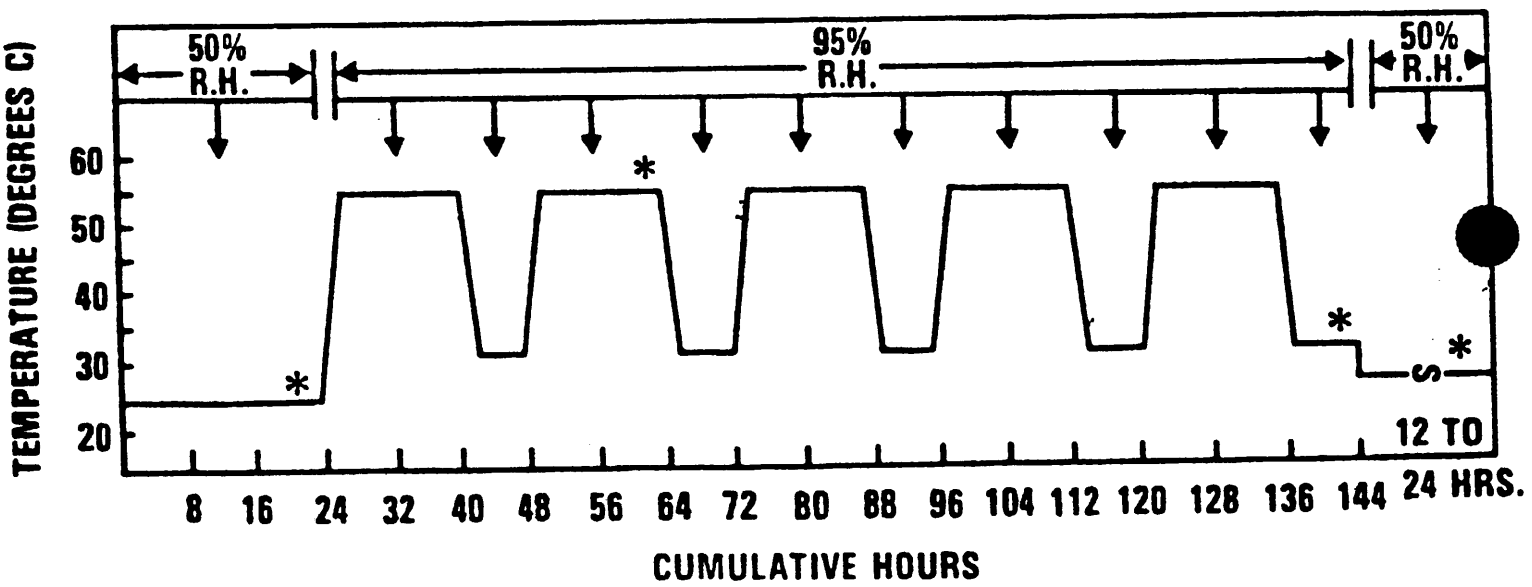
3. Separate Dry Bulb and Wet Bulb Indicators (with suitable resolution) .

The preferred method is (l). If a wet bulb sensor is used, the air flow past the sensor shall be at least 900 feet per minute- Both the dry bulb and wet bulb sensors shall be located in the same air stream with the dry bulb sensor oriented beside or upstream from the wet bulb sensor.

4. 8.1.3.2 Humidity test equipment. The equipment listed in Table X will be required to conduct the humidity test. All measurements shall be made with instruments whose accuracy and calibration has been verified periodically and meet specification requirements.

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* PERFORMANCE MEASUREMENTS

FIGURE 6. Humidity cycle

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TABLE X. Humidity test equipment

Description	Range/Accuracy	Type/Model	Calibration Period
Humidity Chamber and Recorder System	-100 to +275°F/ +20F RF 20-100%	Conrad/FD-8-3-3	As Reg Vd
Oscillosmpe	0-5 MHz 33%	453	As Reqtd
Vertical Amp Plug-In	DC-24 MHz +3%	CA	As Reqtd
Electrical Power	115 VAC, 400 Hz 10, 30A	Facility	N \A
Power Supply DC	O-36 v/5 amps	HP\6266A	As Req'd

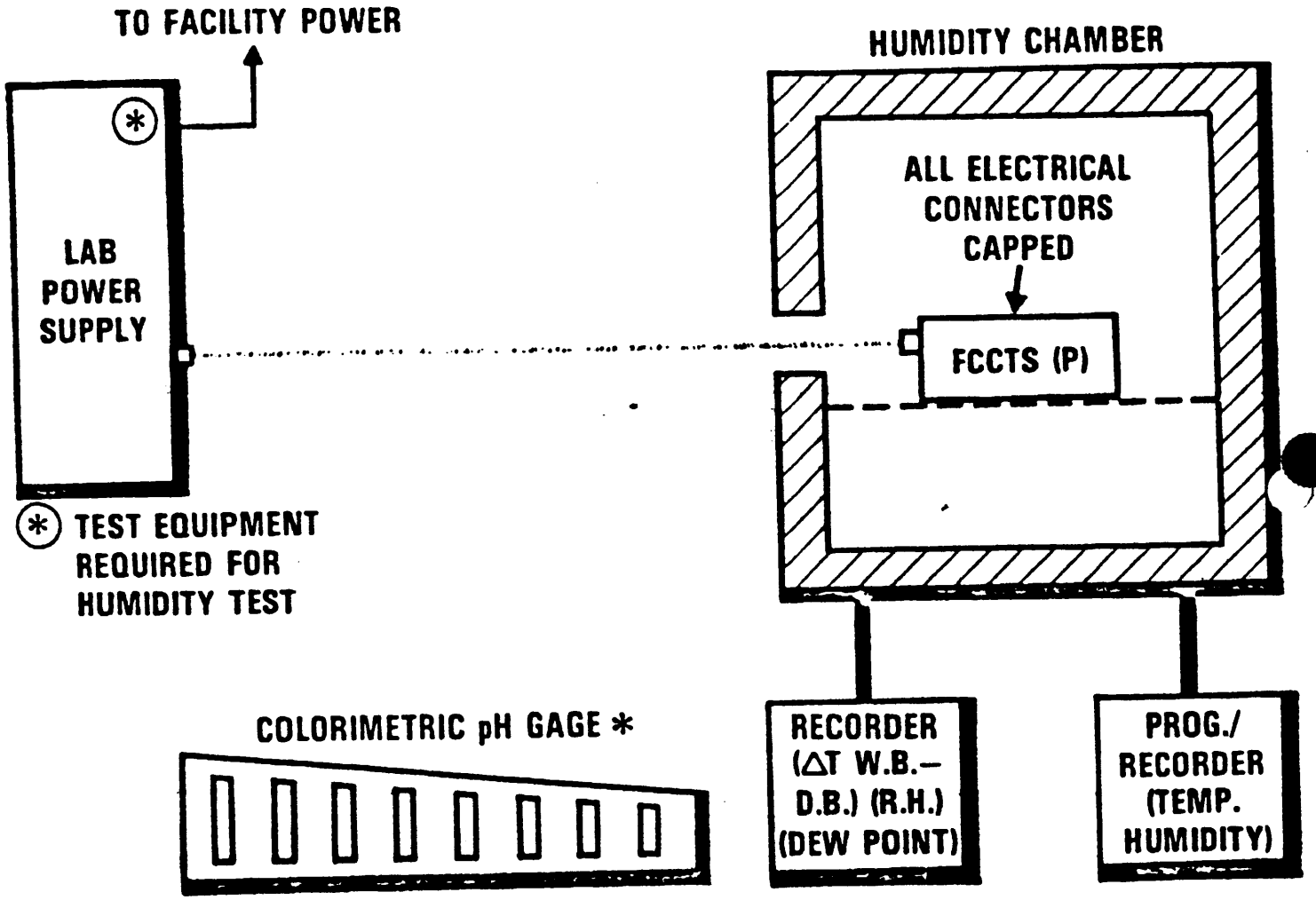
4. 8.1.3.3 Test setup.

4.8.1 .3.3.1 FCC-FTS. The test setup of the FCC-FTS during the humidity cycle shall be such as to simulate actual usage insofar as ractical. The FCC-FTS undergoing test shall be placed in the test e chamber as depicted in Figure 7* Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled the test chamber shall be at standard ambient condition? when the test item is installed. The FCC-FTS shall be oriented within the chamber so that water condensation will not form and collect at any one localized spot.

4. 0.1.3.4 Methods of inspection for humidity tests.

4. 8.1.3.4.1 Test Performance. The chamber conditions for the various steps of the humidity test shall be attained in continuous, uninterrupted cycles and at the rate of change indicated. Thermal and humidity shock shall be avoided at all times and the FCC-FTS under test shall have previously been subjected to the extrae temperatures (-62 and +850C) of the temperature-altitude tests defined in paragraph 4.5.2.1.2 of this procedure.

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= EQUIPMENT REFERENCE NUMBER

NOTE:
* BROMOTHYMAL BLUE pH GAGE USED FOR VERIFYING pH OF WATER USED.

FIGURE 7. Humidity test set-up

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a. with. the FCC-FTS- setup in the humidity chamber in. ..! accordance with paragraph 4.8.1.3.3. 1, and non-operating? close and seal the humidity chamber .

b. Dry the test item at a temperature of not less than 40°C (104°F) nor more than 50°C (122W) for not less than 2 hours.

c. Condition the test item at 250 ± 5oC (770 ± 9°F) and 50 percent relative humidity for 24 hours.

d. Take performance measurements as required on the equipment test data sheets and record the applicable data.

Note: The test item may be readjusted or realigned as necessary to conform to the equipment specification requirements. No further realignment or readjustment shall be permitted throughout the test period other than with accessible controls, external to the test itemt employed for operation of the test item. If repairs, replacement of parts, or adjustments other than by the accessible external controls are made at any time prior to completion of the measurements required at the end of the fifth cycle, all five of the 24-hour cycles shall be repeated; Repairs include any change to the test item that is not made by use of the accessible "controls external to the test item. The test item shall only be operated when specified test measurements are being performed-

e. Subject the test item to five 24-hour cycles in accordance with Figure 5-3. A 24-hour cycle consists of 16 hours at 55° ± 5°C (131° + 9°F) and approximately 8 hours at 300 + 5°C (860 + 9°F)includes transition times). The relatve humidity shall-be maintained at 95 percent,or greater,at both temperatures. Each transition time between 300 ± 50C (860 ± 9°F) and 550 + 5°C (1310 + 9°F) shall be not greater than 1-1/2 hours. 'The relative humidity during each transition need not be controlled. Approximately 2 hours after stabilization during the high temperature and low temperature portions of the first or second cycle, a sampling of the atmosphere in the chamber shall be made to determine that the conditions of temperature and relative humidity are uniform throughout the chamber.

f. Take performance measurements as specified in the equipment test data sheets during the second cycle at 550 + 5oC (1310 ± 9°F) immediately **prior to decreasing to 300 ± 5° (860 ± 9°F.)**.

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The test item shall be **energized only a sufficient time to allow the required warmupd measurements specified in the equipment specification.**

g. After completion of the fifth cycle with the test item in the chamber and the chamber at 30° ± 5°C (860 ± 9oF) and a relative humidity of not less than 95%, take measurements specified in the equipment test data sheets.

h. Condition the test item at 250 ± 5°C (770 ± 9°F) and 50 ± 5 percent relative humidity for not less than 12 hours nor more than 24 hours.

i. While at 25 ± 5°C (770 ± 9°F) and 50 percent relative humidity, take measurements as specified in the equipment test data sheets.

j. **Inspect test item to detect evidence of physical degradation (such as corrosion of metal parts, distortion of plastic parts, and insufficient lubrication of moving parts). when visible corrosion is removed by suitable Cleaning, tbe equipment shall, as a result of the cleaning, be returned to a condition which conforms to the requirements of preoperational inspection and the performance of the equipment shall not be degraded.**

4. 8.1.3.4.2 Post humidity operation. At the completion of the five humidity cycles, the FCC-FTS shall be removed from the humidity chamber and subjected to the post operational performance tests in accordance with paragraph 4S6.2 of this procedure as soon as possible, but not later than one hour after completion of the humidity test. Prior to measurements, excess moisture may be removed frm the exterior surfaces of the FCC-FTS by turning the unit upside down or by wiping **the** external surfaces only. Record the results of this test on the applicable data sheet.

4. 8.1.3.4.3 Moisture test. Following the above check for satisfactory operation the FCC-FTS shall be opened and examined for evidence of entrapped moisture. Record the results of this examination on the applicable data sheet.

4.8.1.4 Vibration. FCC-FTS shall be **subjected to the category (h) and (g) tests required by MIL-STD-810 and MIL-T-28800. The applicable tests** are listed in Table XI-

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4. 8.1.4.1 Test facility.

a. Vibration exciter system. The vibration tests shall be performed on a vibration exciter system capable of shaking the FCC-FTS at the specified amplitudes and frequencies. Transverse action of the exciter (unloaded) shall be less than 20 percent of the input motion and the hUarmonic wave distortion of the fundamental frequency shall not exceed 10 percent.

TABLE XI. Vibration tests

Test Conditions
1. Resonant Search 5-55 Hz (each of 3 axes) non-operating (2 increments)
2. Resonant Dwell (each of 3 axes) non-operating
3. Sinusoidal Cycling 5-55 Hz (each of 3 axes) non-operating :

b. Accelerometer , control. The accelerometer used to measure and control the input acceleration to the test specimen shall be rigidly attached to the base plate of the test specimen vibration fixture, using the stud normally supplied or by cementing in place using Eastman 910 or dental adhesives. The accelerometer shall be positioned as close as possible to the attachment points of the TCC-FTS to the fixture. The frequency response of the control accelerometer should be flat within ± 5%, 5 Hz to 3 KHz at room ambient temperature.

co Test fixture. The holding fixture for the FCC-FTS shall simulate the normal installation as nearly as possible. That is it shall have, 1) provisions for attachment of its base plate directly to the vibration exciter head or the horizontal slip table, and 2) provisions for the rigid attachment of the FCC-FTS to the base plate.

4.8.1.4.2 Vibration test equipment. The test equipment listed in Table XII will be required to conduct the vibration tests. All measurements shall be made with instruments whose accuracy and calibration has been verified periodically and meet specification requirements.

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4. 8.1.4.3 Test setup.

a. FCC-FTS and test station. The test setup of the FCC-FTS shall be such as to simulate equipment transported as secured cargo . The test set cover shall be closed and secure. **The FCC-FTS undergoing test anti the vibration fixture shall be securely attached to the vibration exciter (head or slip table as required) using the** normal mounting attachment points. The test equipment shall be positioned as far away from the vibration exciter as possible to limit the effect of stray magnetic fields. The limiting factors, however, are the length of the interconnecting cables. The FCC-FTS will not be operated during vibration.

be Interconnecting cables. All interconnecting cables from the FCC-FTS to the test station shall be suspended by elastic cord (bungee), positioned at a point approximately six inches from the FCC-FTS termination. All cables, including accelerometer, shall be oriented so that they emanate in a straight line away from the center of the vibration exciter.

c. Accelerometers (equipment). The FCC-FTS under test shall be equipped with accelerometers located on the unit's surface in each of the three perpendicular axes for determining resonance frequencies and acceleration "g".

TAB LE XII. Vibration test equipment

Description	Range/Accuracy	Type/Model	Calibration Period
Random Sine Vibration and Data Acquisition System	5 to 3 KHz, 25 lbs to 150g, 70 in/sec, 1 in. da.	Unholtz-Dickie TA-16400-150	As Req'd
Stop Watch	0-60 secs/mins/+0.1%	Minerva	As Req'd
Oscilloscope	0-5 MHz 53%	Tektronics 453	As Req'd
Vertical Amp Plug in	DC -24MHz 23%	Ca	As Req'd
Electrical Power	115 vat, 60 Hz 10, 30A	Facility	N/A
Electrical Power	115 VAC, 400 Hz 10, 30A .	Facility	N/A

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4. 8.1.4.4 Vibration test procedure.

4. 8.1.4.4.1 Resonance search. Resonant frequencies of the equipment shall be determined by varying the frequency of applied vibration slowly through the specified range at reduced test levels but with sufficient amplitude to excite the item. Sinusoidal resonance search may be performed using the test level and cycling time specified for sinusoidal cycling test, provided the resonance search time is included in the required cycling test time. Vibrate per Table XIII one cycle in each of the three axes.

4.8.1.4.4.2 Resonance dwell. The test item shall be vibrated along each axis at the most severe resonant frequencies determined **in the resonance search. Test levels, frequency ranges, and test times shall be in accordance with Table XIII. If more than four significant resonant frequencies are found for any one axis, the four most severe resonant frequencies shall be chosen for the dwell test. If a change in the resonance frequencies occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain** the peak resonance condition. The final resonant frequency shall be recorded on the Test Data sheet.

TABLE XIII. Resonance dwell

Frequency Increments H z	Vibratory Level (Inches Peak-to-Peak)	Sweep Time per increment* (minutes)	Survey Time per Axis (minutes)	Resonance Dwell (minutes)
5 to 25	0.025 inches (0.635mm)	10	**15**	10
25 to 55	0.020 inches (0.508mm)	5		

*Note: Sweep times specified shall be for a low frequency to high frequency to lot frequency variation for the applicable increment.

4.8.1.4.4.3 Sinusoidal cycling. The test item shall be vibrated thru one cycle along each axis in accordance with the **displacement levels; frequency range (5-55 Hz) and time (15 min.). The accumulated minimum vibration test time per axis shall be 45 minutes. The specified sweep time (15 min.) is that of an ascending sweep only. Record test results on data sheets.**

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4. 8.1.4.4.4 Post vibration operation. At the completion of vibration testing, the post operational tests or paragraph 4. 5.1 data shall be performed. The test results shall be recorded on the sheet.

4. 8.1.5 shock. The FCC-FTS unit shall be subjected to the applicable tests required by MIL-STD-810 and MIL-T-28800. The applicable tests are listed in Table XIV.

TABLE XIV. Shock tests

Test Conditions	Equipment Shock
	2 directions in each of 3 axes (3 shocks each, 15g, half sine) (11 msec nonoperating) (Total of 18 shocks)

4. 8.1.5.1 Test facility. -

a. Shock test system. The shock tests shall be performed on a shock generating system capable of exciting the test specimen to the required waveform, pulse amplitude, and time duration and which was designed and fabricated in accordance with MIL-S-4456. The overall system response ratio, output divided by input (Eo/Ei) shall within the limits specified below:

Frequency Range	Response (Eo/Ei)
2 to 4 Hz	-0.3 @ 2 Hz; -0.1 4 Hz
4 to 2000 Hz	1.0 ± 0.1,
2 KHz to 7 mz	1.0 ± 0.1 1.0 ± 0.1 @ 2 KHz 1.0 KHZ

b. Accelerometer. The accelerometer used to measure the input shock to the FCC-FTS shall be rigidly attached to the baseplate of the FCC-FTS shock fixture, using the stud normally supplied or by cementing in place using Eastman 910 or dental adhesives. The accelerometer shall be positioned as close as possible to the attachment points of the FCC-FTS to the fixture. The accelerometer resonant frequency shall be greater than 14 KHz (30 KHz or higher is recommended) for piezo-electric types. For suitable low frequency response, the accelerometer and load (cathode - follower, amplifier, or other) shall have the following characteristics:

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RC 0.02 5 HZ

Where

R - Load Resistance (ohms)

C - Transducer Capacitance plus shunt capacitance of cable and load (farads)

Strain-gage type transducers may be used provided the undamped natural frequency is equal or greater than 1.5 KHZ with the damping approximately 0.64 to 0.70 of critical. (Resonant frequencies greater than 1.5 KHz are recommended.)

c. Filters. A band-pass filter may be used to display and record the fundamental frequency of the shock pulse providing it has band-pass within the frequency range of 0.2 to 650 Hz and will not degrade the response of the system to exceed the limits specified in item a, above.

d. Fixture. The holding fixture for the FCC-FTS shall be the same test fixture used for the vibration test.

4. 8.1.5.2 Shock test equipment. The test equipment listed in Table XV is required to conduct the shock tests. All measurements shall be made with instruments whose accuracy and calibration has been verified periodically and meet specification requirements.

TABLE XV. Shock test equipment

Description	Range/Accuracy	Type/Model	Calibration Period
Oscilloscope	0-50 MHz\+ 3%	H-P/180-1801A	As Req'd
Mechanical shock and Data Acquisition System	O to 750 g, 100 lbs	LAB/SD-16-42 -100	As Req'd
Scope Camera	N/A	453	As Req'd
Vertical Amp Plug in	DC -24MHz 23%	CA	As Req'd
Electrical Power	115 vat, 60 Hz 10, 30A	Facility	N/A
Electrical Power	115 VAC, 400 Hz 10, 30A	Facility	IV/A

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4. 8.1.5.3 Test setup FCC-FTS. The test setup shall simulate handling, transportation, and service use. The FCC-FTS undergoing test and the shock fixture shall be securely attached to the shock machine carriage using the normal mounting attachment points.

4. 8.1.5.4 Methods of inspection for handling and transportation test.

4. 8.1.5.4.1 Equipment shock. With the FCC-FTS non-operatingr it shall be subjected to three shocks in each of two places in each of the three mutually perpendicular axes. The sequence of test may be in any order.

a. Z' plane

b. Z plane

c. Y plane

d. Y plane

e. X' plane

f. X plane

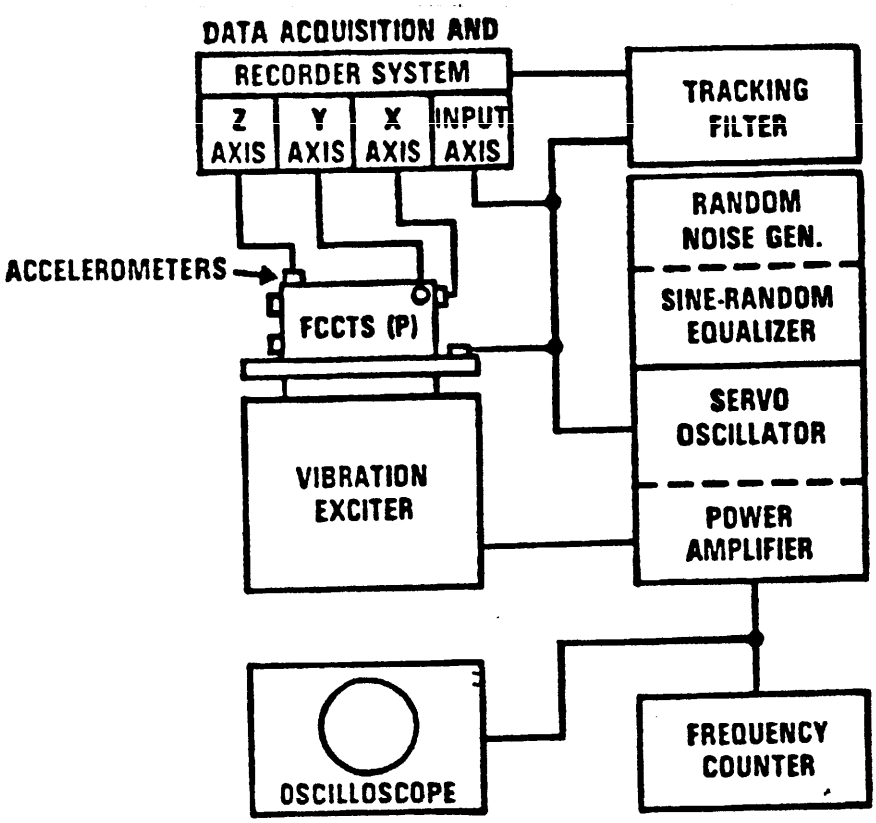
Refer to Figure 8 for axis orientation. The shock pulse characteristics shall be as depicted in Figure 9 and summarized as follows:

Waveform - Half sine
Amplitude -15g ± 3.og
Time Duration - 11 ± 0.77 milliseconds

4. 8.1.5.4.2 Requirements. The shock input spectrum (amplitude~ period, and waveform) shall be monitored for each shock, however only one photograph of the shock spectrum need be made to assure proper system performance and calibration. During the shock test there is no requirement for functional performance. After each series of shocks in each plane, the FCC-FTS mounting attachment points shall be examined for evidence of mechanical failure. After the completion of all 18 shocks (3 axes x 2 planes x 3 shocks each) the FCC-FTS shall be carefully examined for evidence of mechanical failure with all covers open. The test results shall be recorded on the applicable data sheet.

4.8.1.5.5 Post shock oPeration. At completion of test, the test set shall be operated per paragraph 4.5.1 and inspected and results entered on the data sheets.

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AXIS ORIENTATION

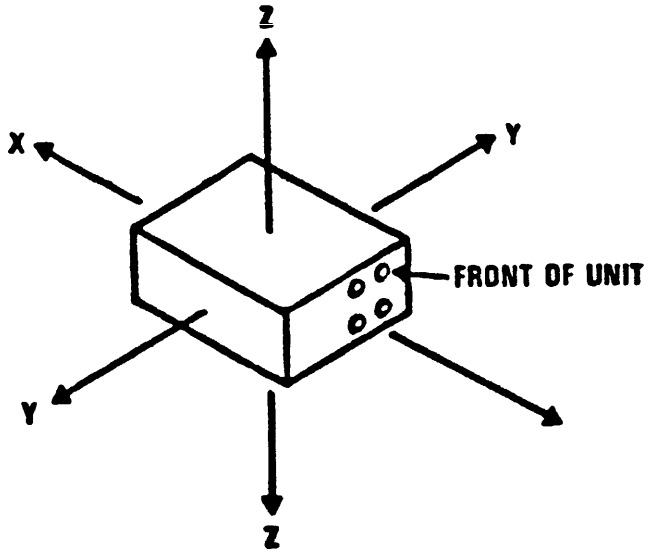
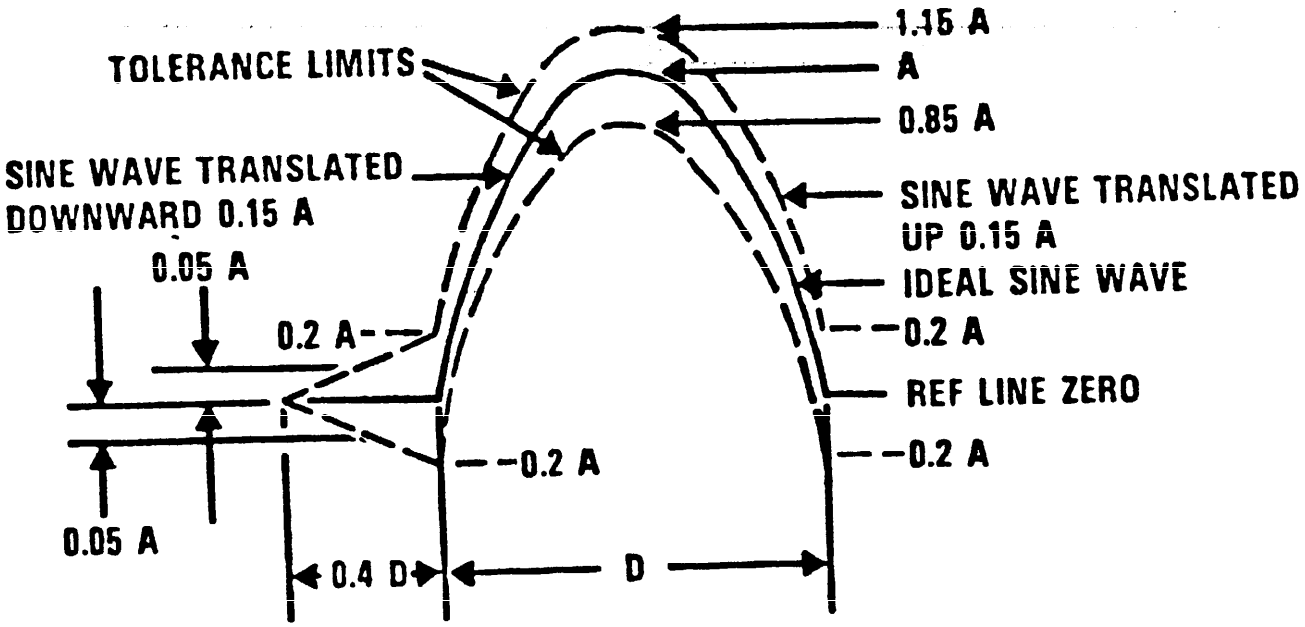


FIGURE 8. Vibration test set-up

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THE OSCILLOGRAM SHALL INCLUDE A TIME ABOUT 3D LONG WITH A PULSE LOCATED APPROXIMATELY IN THE CENTER. THE ACCELERATION AMPLITUDE OF THE IDEAL HALF SINE PULSE IS A, AND ITS DURATION IS D. ANY MEASURED ACCELERATION PULSE WHICH CAN BE CONTAINED BETWEEN THE BROKEN LINE BOUNDARIES IS ACCEPTABLE. THE MEASURED VELOCITY CHANGE (WHICH MAY BE OBTAINED BY INTEGRATION OF THE ACCELERATION PULSE) SHALL BE WITHIN THE LIMITS $V_i \pm 0.1V_i$ WHERE V_i IS THE VELOCITY-CHANGE ASSOCIATED WITH THE IDEAL PULSE WHICH EQUALS 2 AD/TT. THE INTEGRATION TO DETERMINE VELOCITY CHANGE SHALL EXTEND FROM 0.4 D BEFORE THE PULSE TO 0.1 D BEYOND THE PULSE.

FIGURE 9. Half sine shock pulse configuration and its tolerance limits

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4.8.1.6. Bench handling. The FCC-FTS shall be subjected to the applicable tests required by MIL-STD-810 and MIL-T-28800. The applicable tests are listed in Table XIV with cross-reference to MIL-STD-810 and test paragraphs in this procedure.

4. 8.1.6.1 Bench handling test. Bench handling tests should be performed on the FCC-FTS as follows:

The chassis and cover assemblies shall be removed from its enclosure, as for servicing and placed in a suitable position for servicing on a horizontal, solid, wooden bench top at least 1-5/8 inches thick. Using one edge as a pivot, lift the opposite edge of the chassis until one of the following conditions occur (whichever occurs first):

The chassis forms an angle of 45 degrees with the horizontal bench top.

b. The lifted edge of the chassis has been raised 4 inches above the horizontal bench top.

c. The lifted edge Of the chassis is just below the point of perfect balance.

Let the chassis drop back freely to the horizontal bench top. Repeat, using other practical edges of the same horizontal face as pivot points for a total of four drops.

Repeat the above procedure with the test item resting on other practical edges until it has been dropped for a total of four times on each face which the test item could be placed practically during servicing.

TABLE XVI. Bench-handl ing

Test Conditions	MIL-STD-810 Paragraph
4" drop to wood surface	Method 516.2 Procedure V Paragraph 3.7

4.8.1.6.2 Post bench handling inspection/operation. At the conclusion of the test, the FCC-FTS shall be carefully examined for evidence of mechanical-failure. It shall then be reassembled, oonnected, and operated in accordance with the post operational tests of paragraph 4.5.1. The results shall be recorded on the applicable data sheets.

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4.8.17 Watertight test. The FCC-FTS unit shall be subjected to a watertight test upon completion **of the** transit **drop** test **(paragraph 5. 8). The applicable test is listed in Table XVII with cross reference to MIL-T-28800 and the test paragraph in this procedure.**

4. 8.1.7.1 Test facility. The watertight test shall be performed by submersion of the FCC-FTS within a container of sufficient capacity to allow 3 feet of tap water above the FCC-FTS. Alternately the FCC-FTS case shall be covered with 3 inches of water and the pressure within the FCC-FTS reduced to 13.3 psi (approx 2500 ft altitude, 685 torr, 27 in Hg).

4. 8.1.7.2 Test performance. With the FCC-FTS covers closed and secured, place the non-operative test unit in the watertight test container . Using tap water, fill the oontainer until a depth of 3 feet above the test unit is obtained. Alternately fill the container to a depth of 3 inches above the test unit and reduce the pressure within the case to 13.3 ± 1 psi. Maintain these conditions for a minimum period of one hour.

4.8.1.7.3 Post watertight test. Upon completion of the **watertight test,** remove the FCC-FTS unit from the container and remove any excess water from the case surface by allowing it to run off or gently removing by wiping with industrial Kimwipes or equivalent. Open the test unit and examine for evidence of water or moisture within the FCC-FTS case. Entry of water during the test shall be cause for rejection. Record the results on the applicable data sheet. Obtain a test equipment list and a glossy photograph **of** the test setup.

TABLE XVII. Watertight

Test conditions	MIL-T-28800 paragraph
Water submersion 3 feet (1.3 psi), 1 hour	3.9.6.1

4.8.1.8 Transit dr**op.** The **FCC-FTS shall be subjected to the applicable tests required by MIL-T-28800, which invokes MIL-STD-810, Method 516.2, Procedure II.**

4.8.1.8.1 Transit drop test. The transit drop test shall be performed *on* **the test set as** follows:

The combination case including a full compliment of equipment (i.e., panel assembly, logic assembly, cables, manual) shall be placed in the case and the cover shall be locked in place.

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b. The test set shall be dropped once on each corner using a quick-release hook from a height of 12 inches onto 2-inch plywood backed concrete (8 total).

c. Repeat for each edge (12 total).

d. Repeat for each face (6 total)

for a total of 26 drops.

4.8.1.8.2 Post drop inspection/operation. Upon completion of the transit drops, the test set shall be operated per paragraph 4.5.1 and inspected. There shall be no evidence of damage to the case, such as openings along any joint, loosening of partitions, splitting or cracking of case material. Any partial or total opening of catches shall be cause for rejection of the case. The case and cover shall function in the same manner as prior to the drop tests. Record the results on data sheets.

TABLE XVIII Transit drop

Test conditions
Transit Drop from 12" for total of 26 drops

4.8.1.9 Salt atmosphere. The FCC-FTS shall be subjected to the test Method 509.1 of MIL-STD-810. Operation of the FCC-FTs shall not be required during testing and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as visible degradation of the equipment surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish of metal surface.

4.8.1.10 Fungus. The FCC-FTS shall be subjected to the test of Method 508.1, Procedure I, of MIL-STD-810. Gasket sealed or other non-hermetic sealed test items shall be opened during test exposure and all internal and external surfaces shall be sprayed with spore suspension. Operation of the equipment is not required during testing, but examination of the concentration after exposure period shall reveal no evidence of viable fungus on any surface.

4.8.1.11 Explosive Atmosphere. The FCC-FTS shall be subjected to test Method 511.1 procedure II, of MIL-STD-810, except test shall be limited to a maximum altitude of 20,000 feet. At the conclusion of this test, subject the system to the performance tests of Table II.

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4.8.1.12 Electromagnetic radiation. The FCC-FTS shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL- STD-461 for the requirements of 3.2.1.

4.8.1.13 Sand and dust. The FCC-FTS shall be subjected to test Method 510.1, Procedure I, of MIL-STD-810, except that the air velocity in Steps 1, 2, and 3 shall be 200 ± 100 FPM. Step 3 may proceed immediately after stabilization in Step 2, if desired. At the conclusion of this test, subject the FCC-FTS to the performance tests of Table II.

4.8.1.14 Preconditioning (burn-in). Subject each FCC-FTS to the burn-in of Appendix B.

5. PACKAGING

5.1 Preservation, packing, packaging, and marking. The FCC-FTS shall be preserved, packaged, packed and marked in accordance with the retirements of the preservation and packaging data sheet at the level of protection specified in the contract.

5.2 Commercial packaging. Commercial packaging shall be in accordance with ASTM-D3951.

60 NOTES

6.1 Intended use. This equipment is intended for the functional test and fault isolation of the Fire Control Computer, M22.

6.2 ordering data

6.2.1 Acquisition requirement. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable Packaging Data Sheet.

c. Selection for applicable levels of preservation, packaging~ and packing.

d. Applicable stock number.

e. First article inspection requirements.

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6.3 **Previous revisions.** **Previous revision of MIL-F-48673 were for the US Army Armament Research and Development Center internal use Only. and were not forwarded to the Naval Publications and Form Center for publication.**

6.4 **Subject term (key word) listing**

Amendments
Fire Control Computer
Field Test Set
Military Specifications
Technical Manual
Control Interface Siqnals
Test Points
Control Panel
Test Results Display
Maior Component List
Preconditioning (burn-in)
Operational Test
Test Set-Up
Calibration
Test Equipment

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1270 -A057)

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APPENDIX A

CALIBRATION PROCEDURE

10. S COPE

10.1 scope. This appendix provides instructions for calibration procedures, the required test equipment, and burn-in requirements to be applied to all deliverable systems. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

20. **APPLICABLE DOCUMENTS.**

20.1 Government documents. The following documents form a part of this appendix to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-731 - Reliability Design Qualification and Production Acceptance Tests

30. REQUIREMENTS

30.1 Requirements. The Section 3 specified requirements appropriately cross referenced to the tests contained in Section 4.

40. CALIBRATION PROCEDURE

40.1 Test Equipment. **Required test equipment is listed in Table XIX.**

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TABLE XIX. Test equipment

Nomenclature	Minimum Use Specifications	Suggested Equipment
D.C. Power Supplies (2 required)	Range: +13.5 to +16.5 VDC at 1 amp, output isolated from power ground; -13.5 to -16.5 VDC at 1 amp, output isolated from power ground Accuracy: Set with DMM	Power Designs 3650S (2 required)
DMM	DC: Range : +4.9 to +16.5 VDC Accuracy: 0.5 percent AC: Range: 9.9 to 10.1 VAC Accuracy: 0.25 percent	Fluke 8600A
Electronic Counter	Range: 388 to 412 Hz Accuracy: 1 percent	Fluke 1900A
Decade Resistor	Range: 10 Kohms at 10 ma Accuracy: 1 percent	Phipps and Bird 236A

40.2 PRELIMINARY PROCEDURE

40.2.1 Step 1. Visually inspect FCC-FTS for any physical defects. If any are noted, corrective action shall be taken before continuing calibration.

40.2.2 Step 2. Carefully remove test set panel from carrying case .

40.2.3 Step Connect test set power cord to test set J4 connector and to an appropriate 28 VDC, 10 amp source

Verify that POWER indicator is extinguished. If not, depress the extinguish.

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40.3 CALIBRATION PROCEDURE

40. 3.1 **Internal +5 VDC power supply, overvoltage protective measurement.**

40. 3.1.1 Step 1. Set external DC power supply for minimum output and to current limit at 100 ma.

40.3.1.2 Step 2. Disconnect internal power lead at terminal strip TB2-1.

40. 3.1.3 Step 3. Connect test circuit of Figure 10.

40. 3.1.4 Step 4. Slowly increase output of external DC power supply (do not exceed +6.5 VDC) until crowbar condition occurs. Just prior to crowbar, the DMM shall indicate +5.9 to +6.1 VDC. Adjust test set OVP as required.

40.3.1.5 Step 5. Disconnect test circuit of Figure 10 from test set.

40. 3.1.6 Step 6. Reconnect power supply lead removed in Step 2.

40.3.2 +5 VDC output.

40.3.2.1 step 1. **Depress test set** POWER **switch. It shall be illuminated .**

40. 3.2.2 Step 2. Connect DMM to test set terminal strip TB2-1 (+) and TB2-2 (-).

40. 3.2.3 Step 3. The DMM shall indicate +4.9 to +5.1 VDC. Adjust test set power supply as required.

40. 3.2.4 Step 4, **Disconnect** DMM from test set.

40.3.3 Reference oscillator measurements.

40.3.3.1 Step 1, Set decade resistor to 10K ohms.

40. 3.3.2 Step 2, Set **both external DC power supplies to +13.5 to +16.5 VDC and set power switches to off.**

40.3. 3.3 Step 3, Connect test circuit of Figure 11.

40.3.3.4 **Step 4,** Simultaneously set both external power supply power switches to on.

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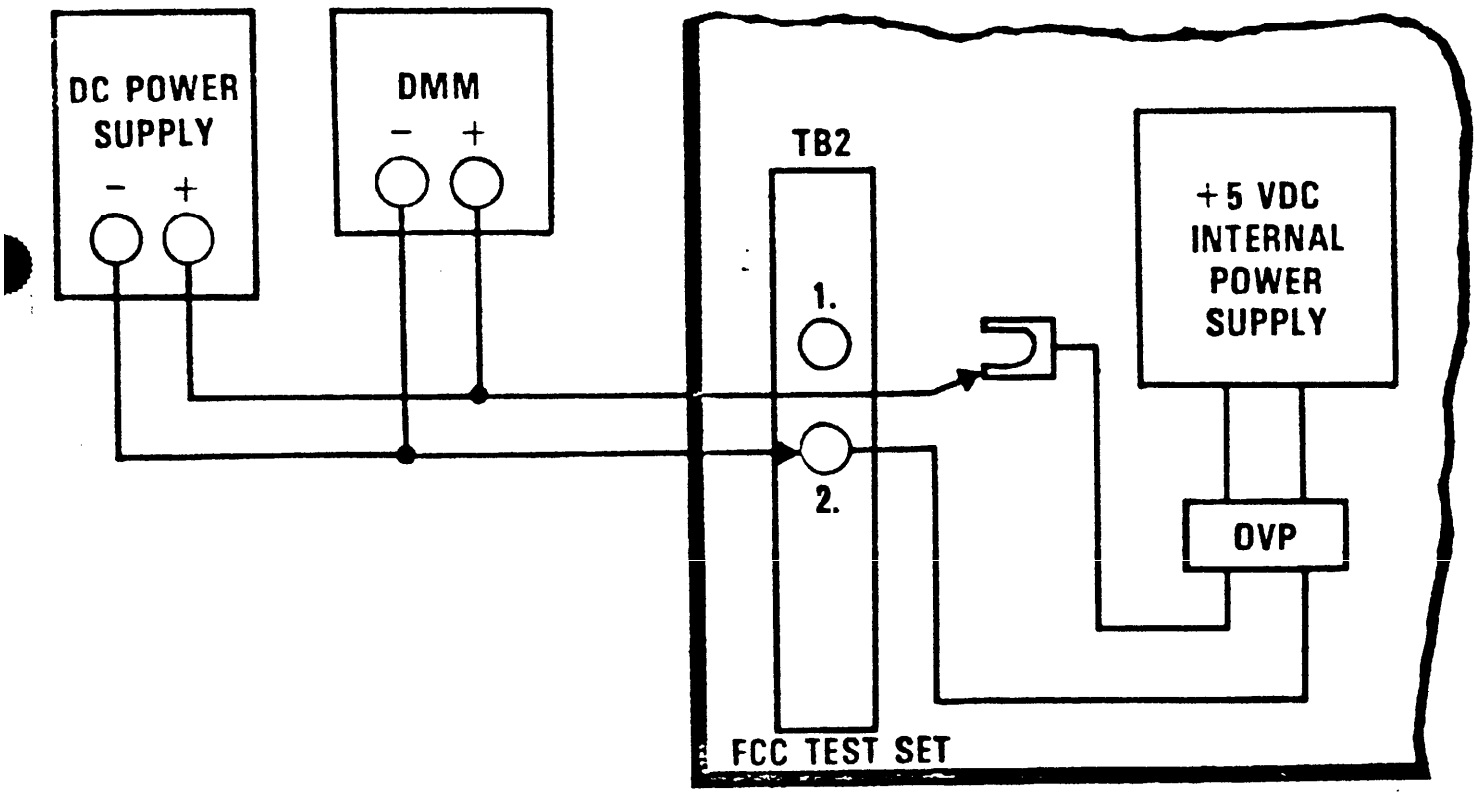


FIGURE 10. OVP measurement set-up

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40. 3.3.5 Step 5. **The DMM shall indicate 9.9 to 10.1 VAC. Adjust oscillator board R2 as required.**

40.3. 3.6 Step 6. The *electronic* counter shall indicate 388 to 412 Hz. Adjust oscillator **board R4 as required.**

40. 3.3.7 Step 7. Simultaneously set both external power supply switches to off.

40. 3.3.8 Step 8. Disconnect **test** circuit of Figure 11 from test set.

40. 3.3.9 **Step** 9. Depress test set POWER switch. It shall be extinguished .

40.3.3.10 Step 10. **Carefully replace test set panel in carrying case.**

40. 3.4 SUPPLEMENTARY INSTRUCTIONS.

40.3. 4.1 Step **Seal unit in** such a manner that unauthorized entry may be readily detected.

40. 3.4.2 Step 2. Affix a calibration decal to the COBRA FCC-FTS that reflects the calibration status and interval defined in the Calibration Control System.

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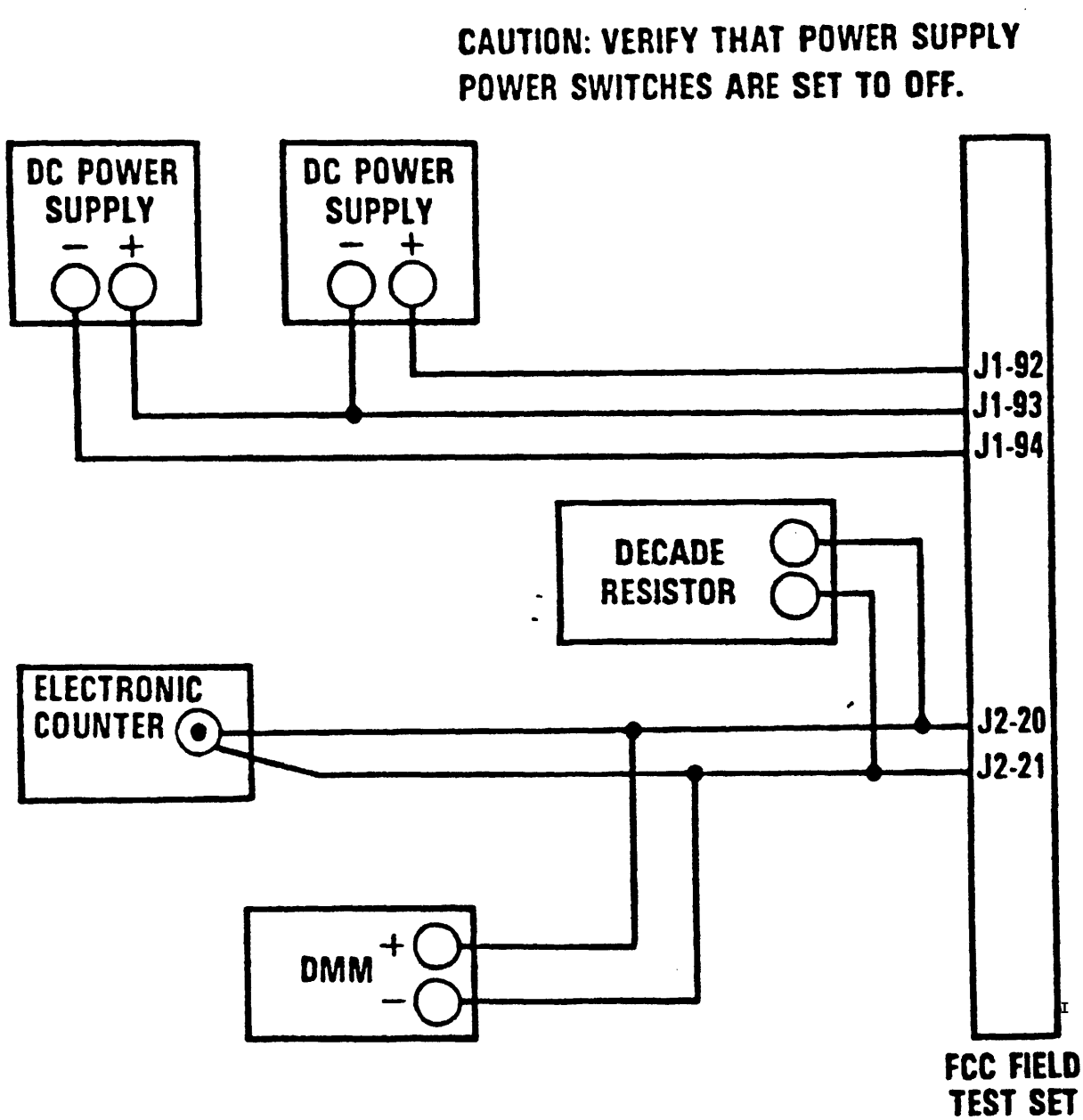


FIGURE 11. Reference oscillator measurement set-w

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APPENDIX B

50. BURY-IN REQUIREMeNTS

50.1 Burn-in. Burn-in is to be applied to all deliverable systems .

50.2 Test environments. Environmental exposure for the FCC-FTS during burn-in is Level F of MIL-STD-781. Cycle time shall be determined by thermal survey. **Chamber temperature change shall be programmed equal to or greater than 5° per minute. Figure 3 illustrates one cycle of the environmental profile described below.**

Temperature	**-54°C to +71°C (-65°F to 160°F)**
Temperature Cycling	**Temperature cycling shall be time to stabilize at low temperature followed by time to stabilize at the high temperature,** plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at 30 ± 5 Hz-measured at the mounting points. **The duration of vibration shall be at least 10 minutes during each hour of equipment operation time.**
Equipment On-Off Cycling	**Equipment off during cooling cycle and on during heating cycle.**
Input Voltage	**Specified voltage +5 -2%.**
Input Voltage Cycling	**a. Nominal voltage for first 1/2 of "ON" time starting at the beginning of the heating period.** **b. Momentary off.** **c. 100% of nominal for next 1/3 of "ON" time.** **d. Momentary off.** **e. 90% of nominal** for next **1/3 of "ON" time.**

50.3 Number of Cycles.

50.3.1 Minimum number. The minimum number of cycles shall be five (5) with the last two (2) cycles failure free.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-48673 (AR)	FIRE CONTROL COMPUTER, FIELD TEST SET (FCC-FTS)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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